



Growth and development

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Room 215

Why does human life history evolve?

What do we have to explain?

Childhood

- **Early weaning** → Compared to other primates
- **Steady rate of body growth**
- **High rate of brain growth**



Adolescence

- **High rate of growth**
- **Growth spurt**

Postreproductive life

- Life is about turning energy into offspring...

Generally longer & slower life

- Extended period to allow brain to grow (necessary because larger brain *and* born with relatively smaller brain) → Born @ early stage of development but need to obtain a much larger brain size
- Childhood allows mothers to offset own long growth period by having multiple dependent offspring
 - Cost is offset through alloparenting
- Childhood allows child to learn

Key

Take-home message:

Evolution of childhood necessitates a very complex set of changes – to the individual but also the community it grows up in

weaned but still dependent for food

Why does human life history evolve?

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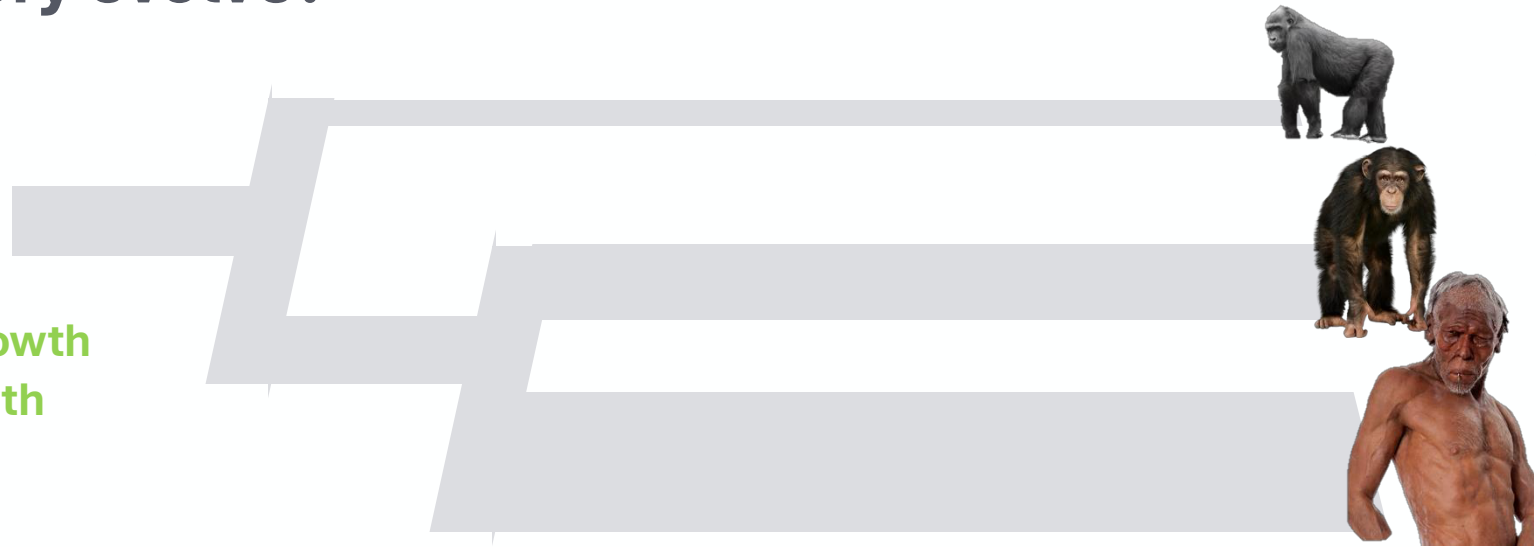
Adolescence

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You help take care of monkeys

- Learning (but why that growth spurt)?
- Cooperative breeding (but why that growth spurt)?
- 'Safe' sexual practice (look mature but not as fertile as adults)?
- Catch-up of skeletal growth: earlier phases marked by **brain** growth, not somatic growth

Explains why growth spurt was selected for

Key

Non-mutually exclusive hypotheses for the evolution of adolescence – but these are hard to test

When does human life history evolve?

Childhood

- Early weaning
- Steady rate of body growth
- High rate of brain growth

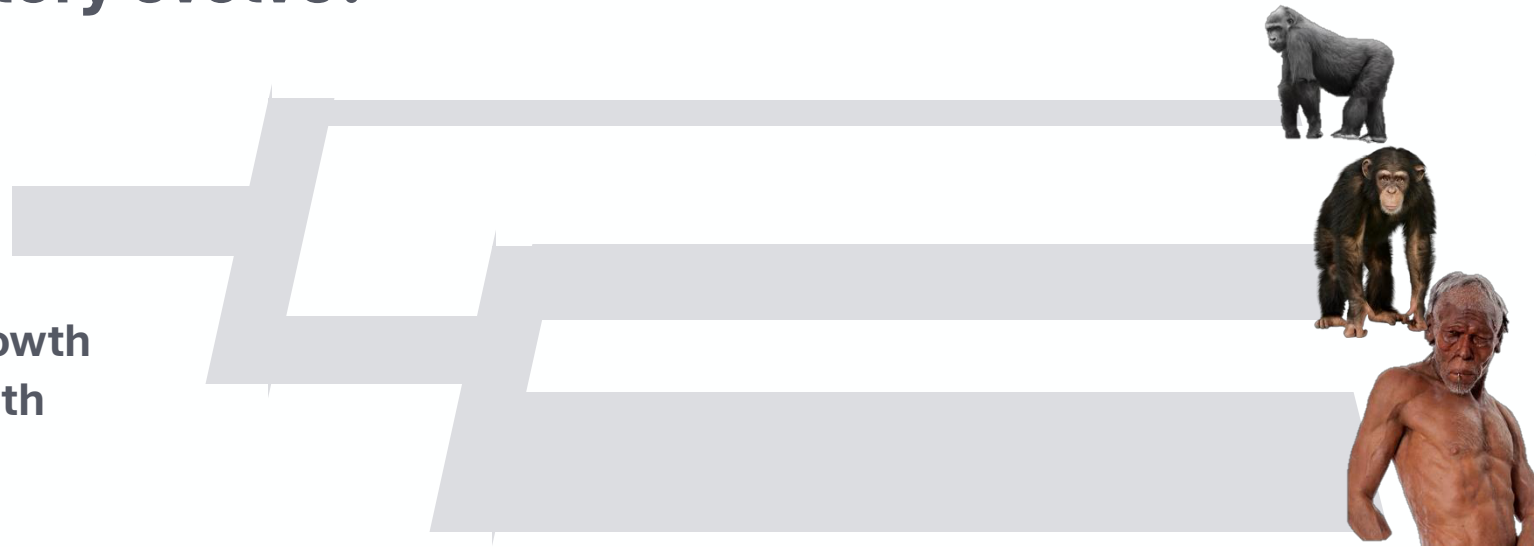
Adolescence

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Postreproductive life

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Generally longer & slower life



When does human life history evolve?

Comparative method
to see when the
stages evolved

Childhood

- Early weaning
- Steady rate of body growth
- High rate of brain growth



Adolescence

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Generally longer & slower life

When does human life history evolve?

Childhood

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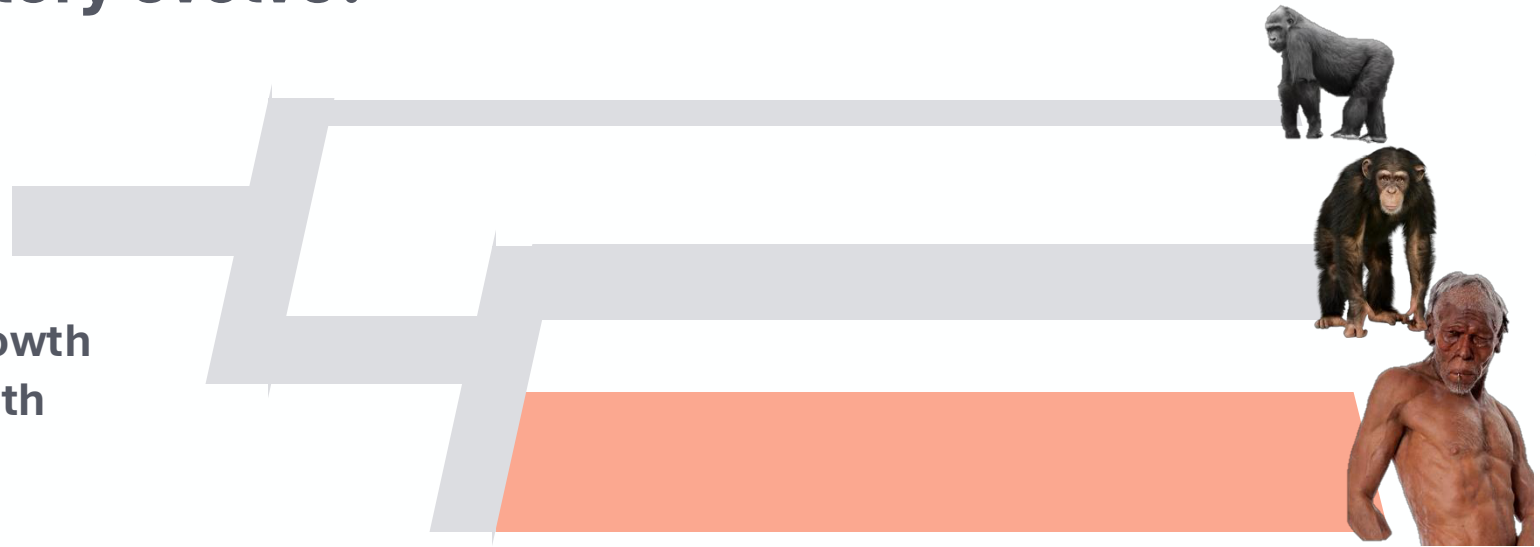
Adolescence

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Postreproductive life

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Generally longer & **slower** life



When does human life history evolve?

Childhood

- Early weaning
- Steady **rate** of body growth
- High rate of brain growth

We need to think abt rates of growth - which you can see in fossils



Adolescence

- High **rate** of growth
- Growth spurt

Postreproductive life

- Life is about turning energy into offspring...

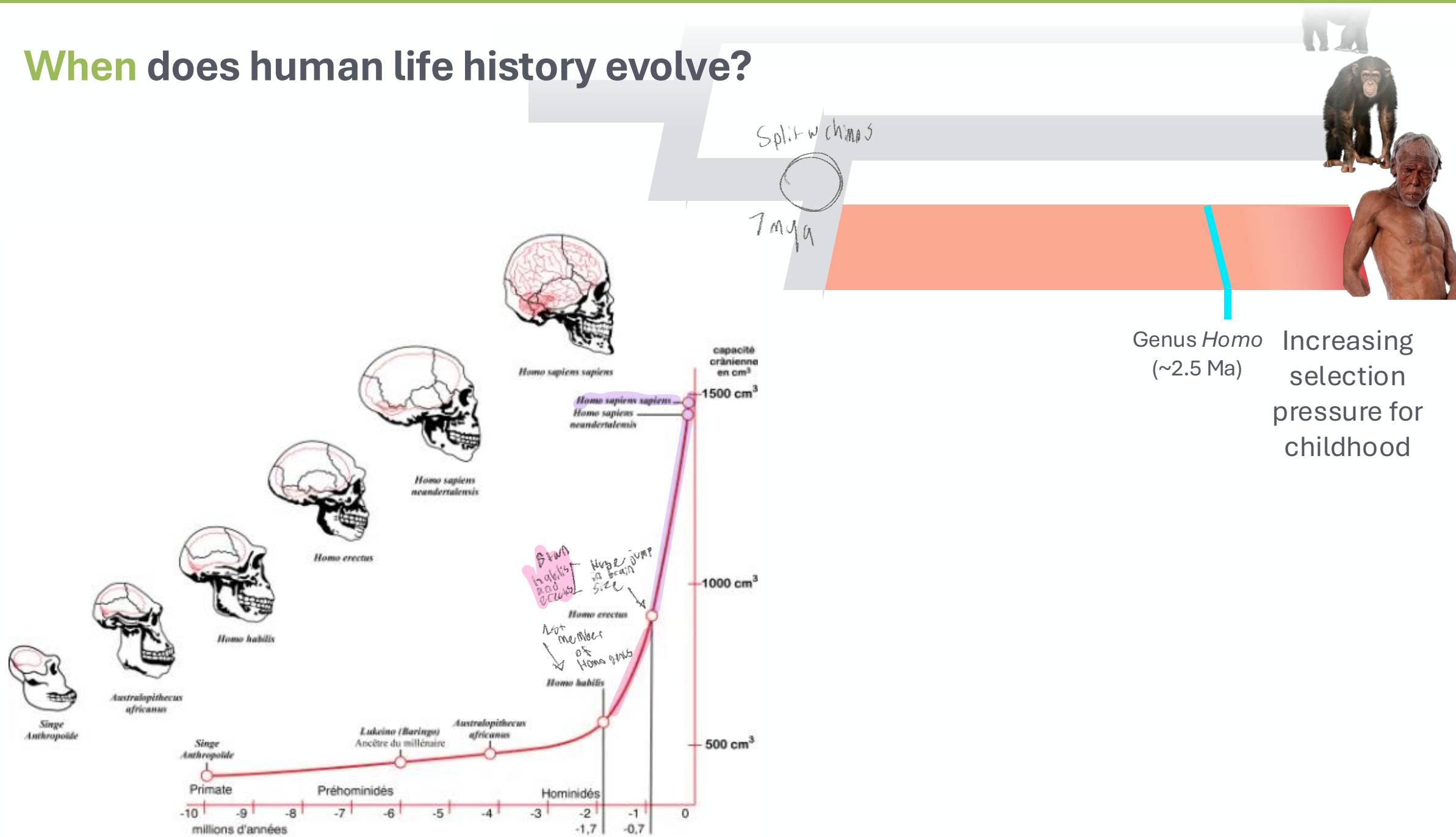
Generally longer & **slower** life

+ signals of cooperation

+ brain size

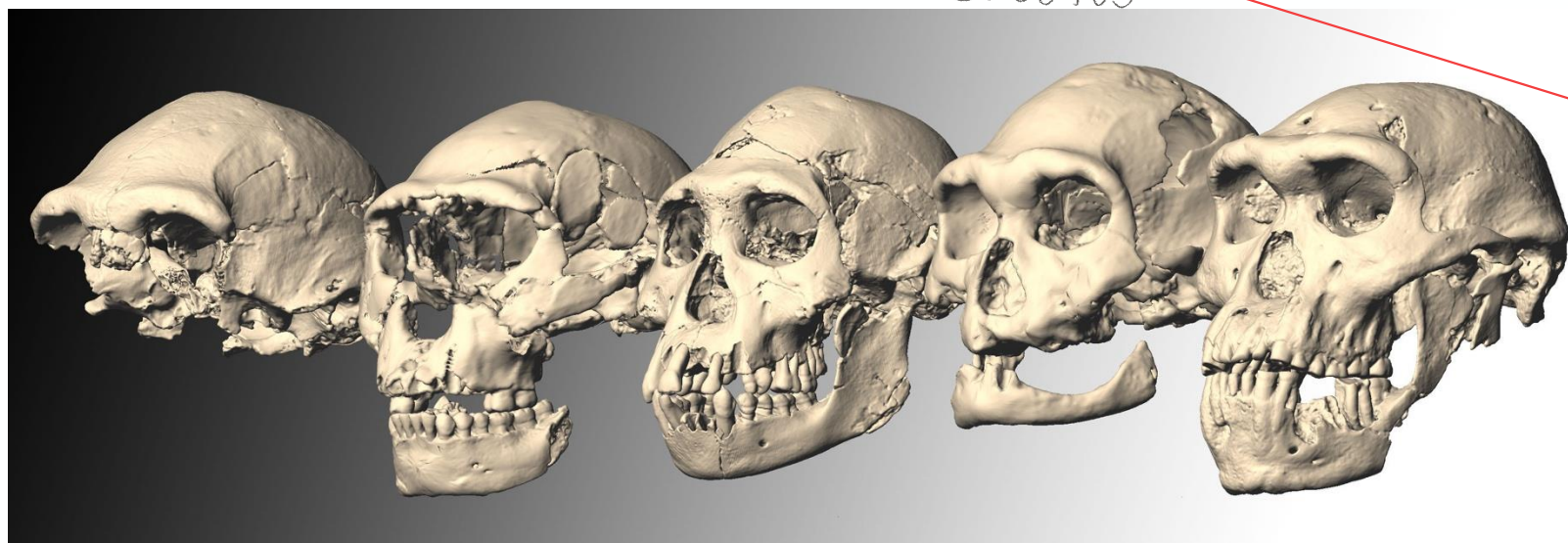
Can be found in fossils and food sharing would be indicative of childhood

When does human life history evolve?





When does human life history evolve?



Erectus

Genus *Homo*
(~2.5 Ma)

Increasing
selection
pressure for
childhood

Cooperation:
Dmanisi?
(~1.8 Ma) Food
sharing

One
of these are
very old and
had no teeth ->
was able to survive w/o
teeth -> cooperation via food
sharing

When does human life history evolve?

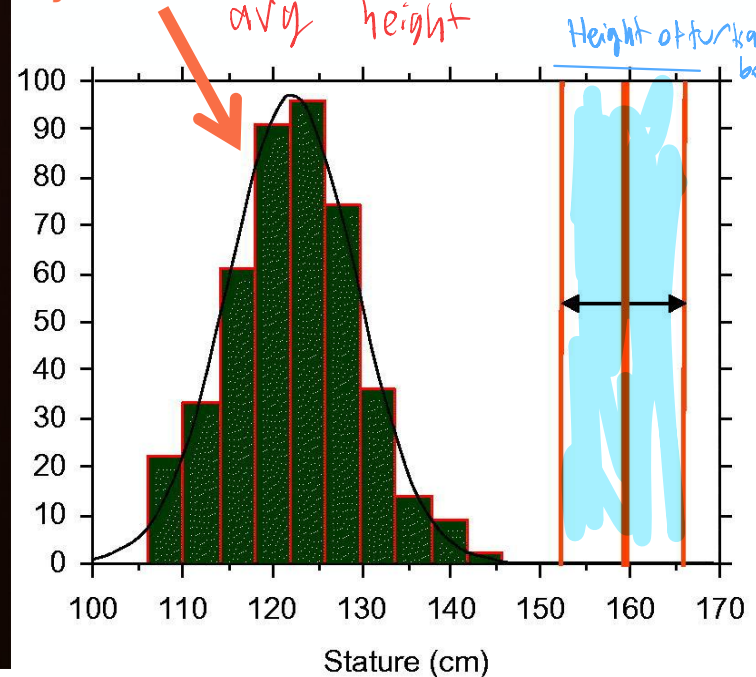


Turkana boy
Homo erectus
~1.8 Ma

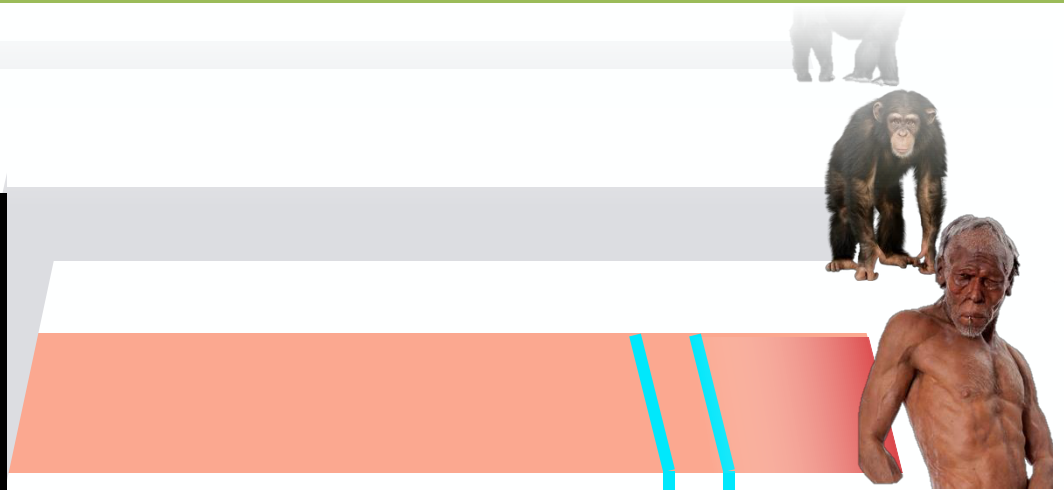
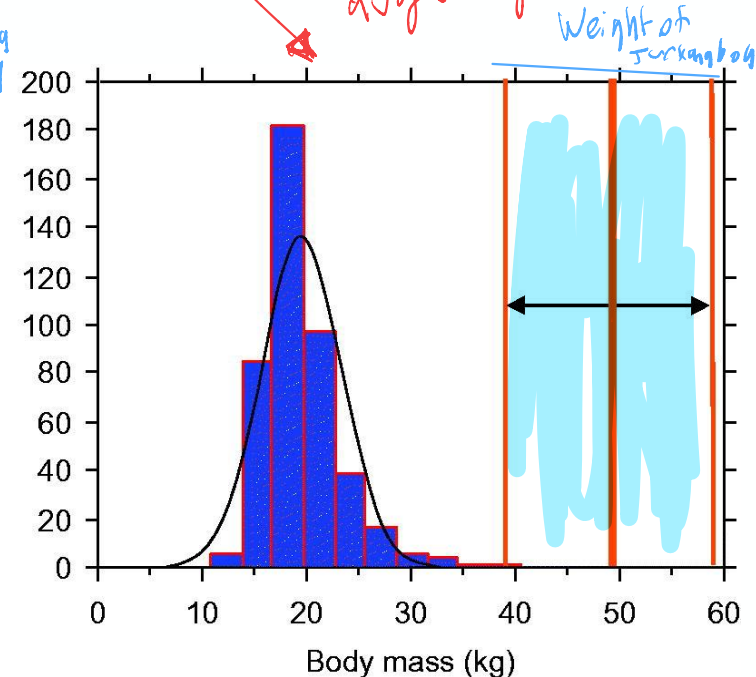


Teeth suggest: 8 years old

Human 8-year-olds



Human 8yo avg weight



Genus *Homo*
(~2.5 Ma)

Increasing selection pressure for childhood

Cooperation:
Dmanisi?
(~1.8 Ma)

When does human life history evolve?

Turkana boy
Homo erectus
~1.8 Ma

8 years of life, but had attained height
& weight of human 15-year-old

No evidence for slower human life
history at 1.8 Ma

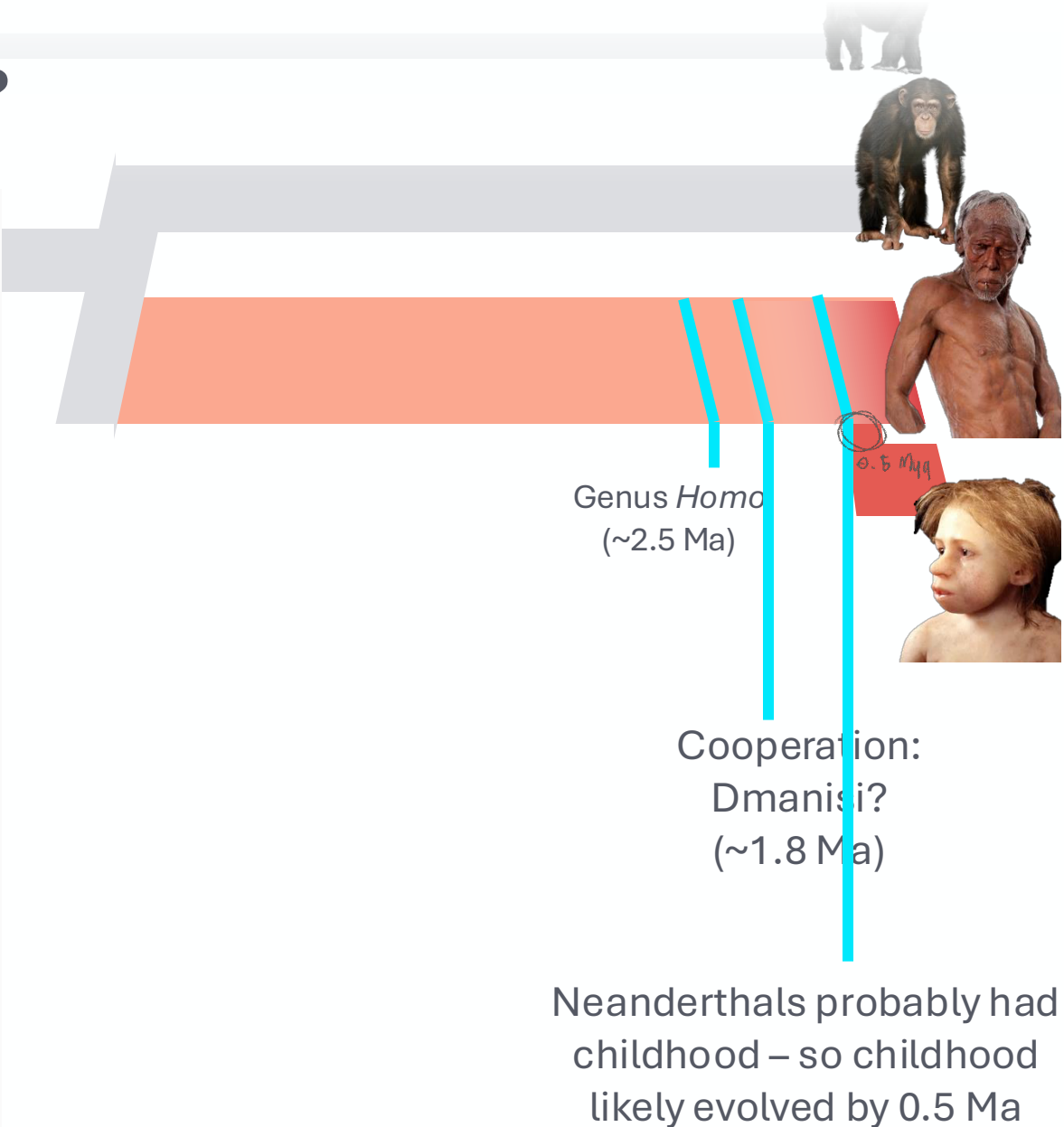
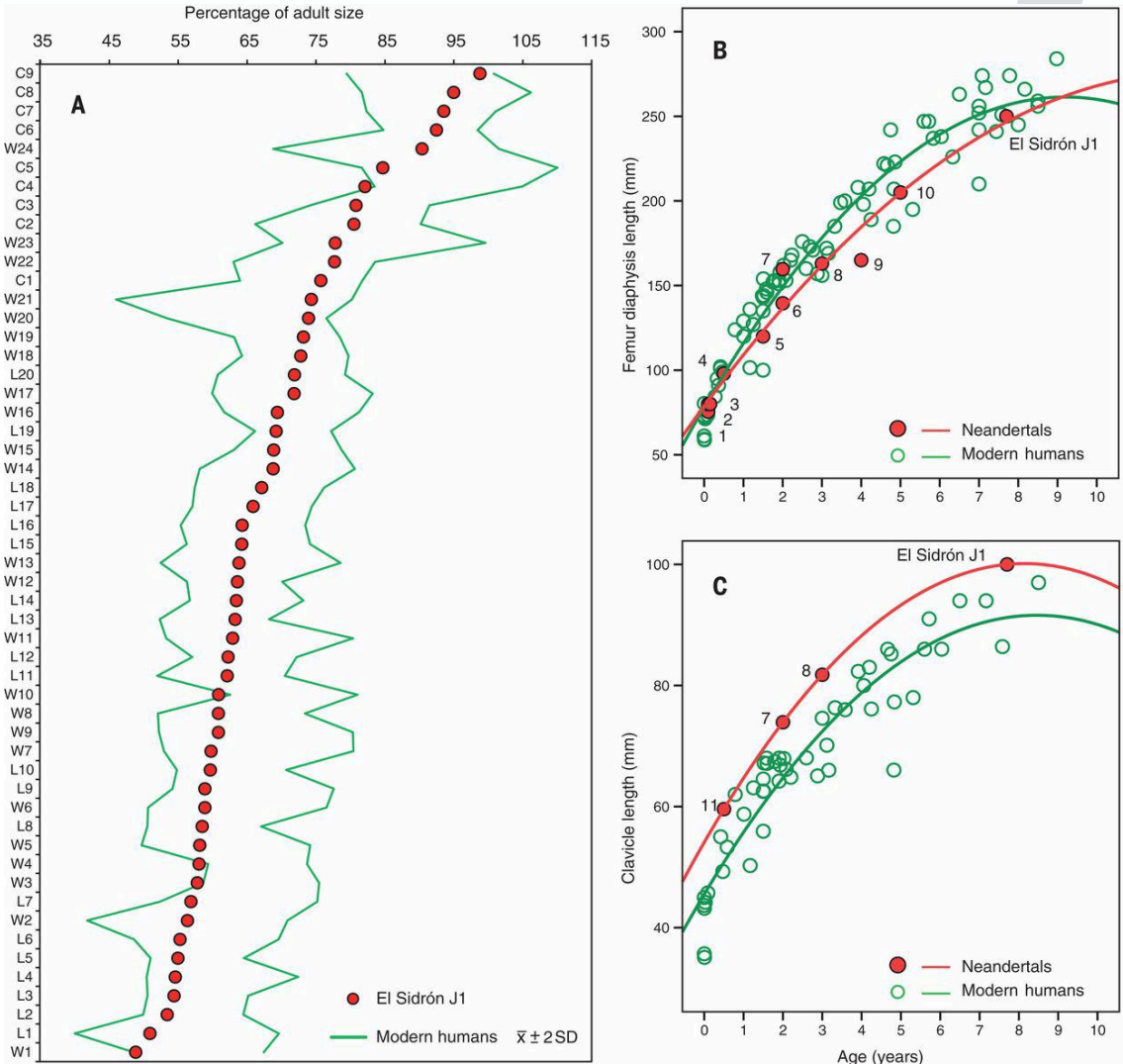
Genus *Homo*
(~2.5 Ma)

Increasing
selection
pressure for
childhood

Cooperation:
Dmanisi?
(~1.8 Ma)



When does human life history evolve?



Evolution

Framework

Pre-, perinatal

Childhood & juv.

Adol.

The big take-home messages

- What?
 - Life history is a species-specific pattern of energy allocation to biological functions
 - These functions can be roughly divided into those relating to growth, those relating to maintenance, and those relating to reproduction
 - Life history stages are characterized by distinct allocation patterns
 - General mammalian life history is somewhat tinkered with in primates (M1 erupts before weaning, long life); this primate pattern is completely broken in humans
 - Human life history has three unique stages: childhood, adolescence, and a postreproductive period
- Why?
 - Hard to say – many non-mutually exclusive explanations. However, likely very much due to our large brains!
- When?
 - Between 1.8 and 0.5 million years ago

What are we talking about?

Definitions

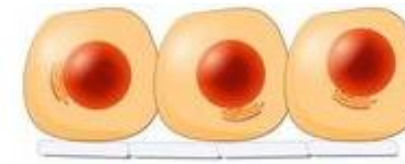
1. **Growth:** quantitative increase in size or mass

- Hyperplasia → cell # increases, size of cells is the same
- Hypertrophy → cell # is the same, grows in size

2. **Development:** progression of changes, either quantitative or qualitative, that lead from an undifferentiated or immature state to a highly organized, specialized, and mature state

↳ Incl. cell differentiation, but we are thinking abt organism as a whole, not cellular level

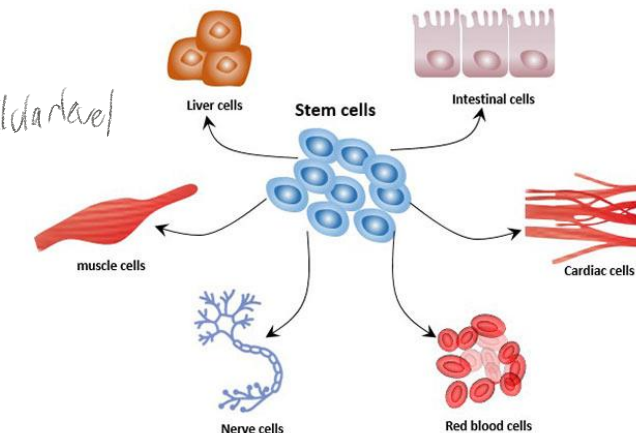
3. **Maturity:** fully functionally capable



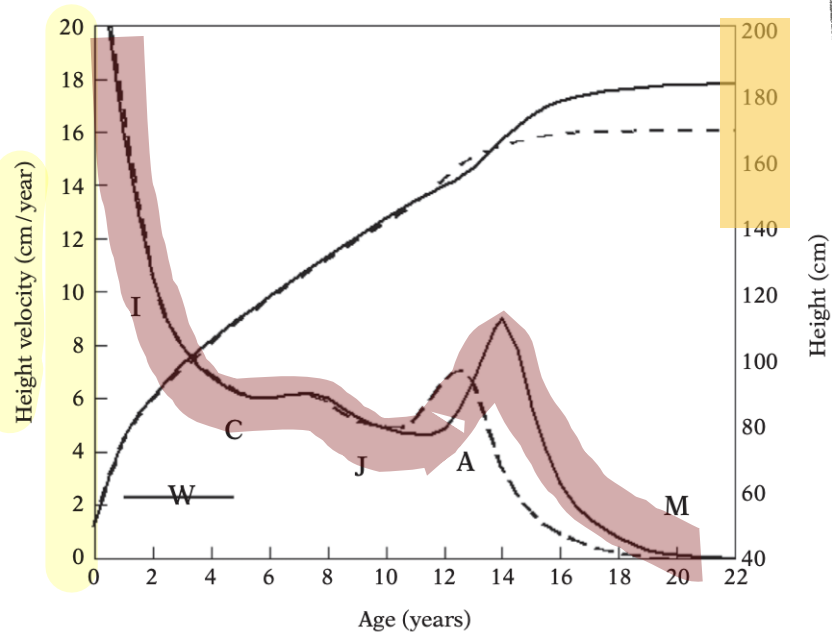
Hypertrophy



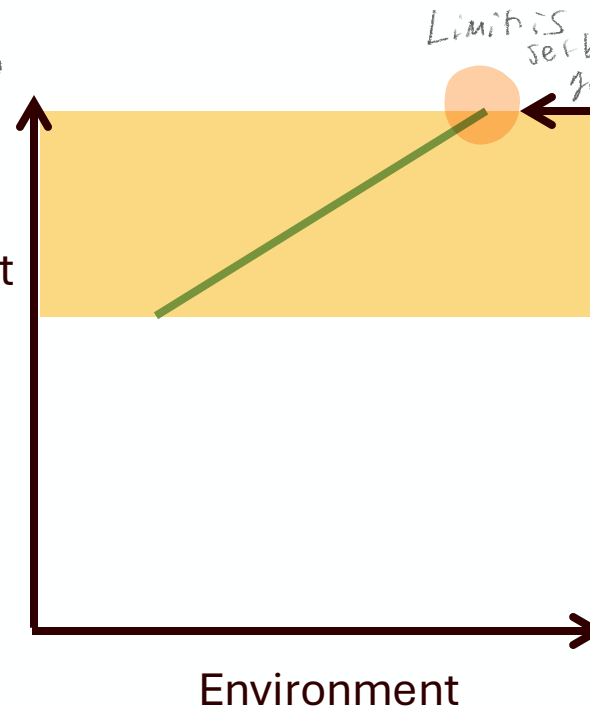
Hyperplasia



Today's framework

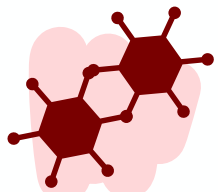


End goals
adult
weight +
Adult
height



End point of height

is not
set at birth
↳ A single genome can attain different
adult height - plasticity, variance in height
by environment

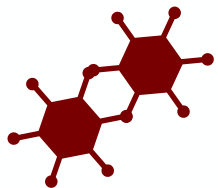
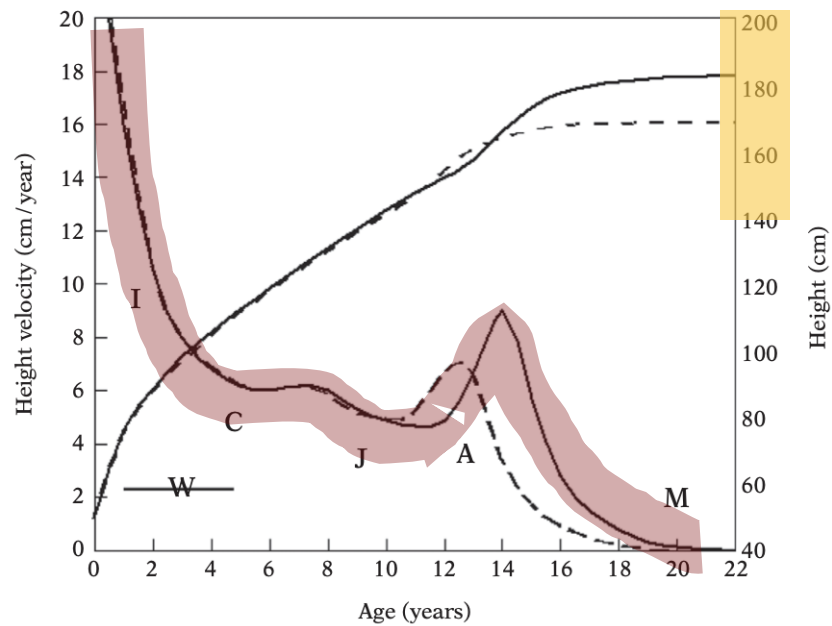


Hormonal pathways
control growth



And these reflect long-
term genetic adaptation
(in many cases preceding
the hominin lineage)

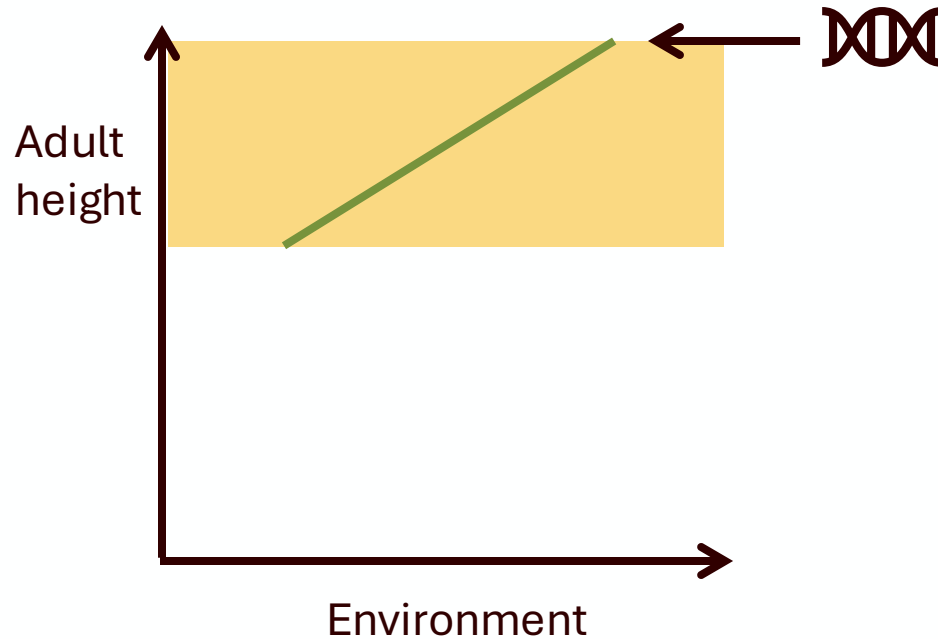
Today's framework



Hormonal pathways
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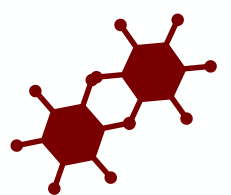
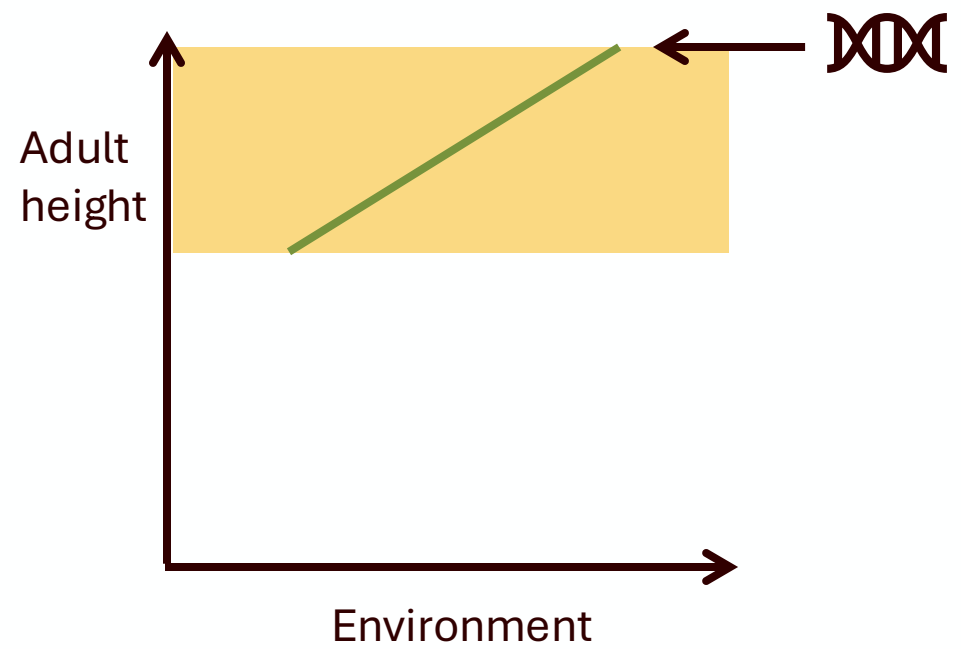
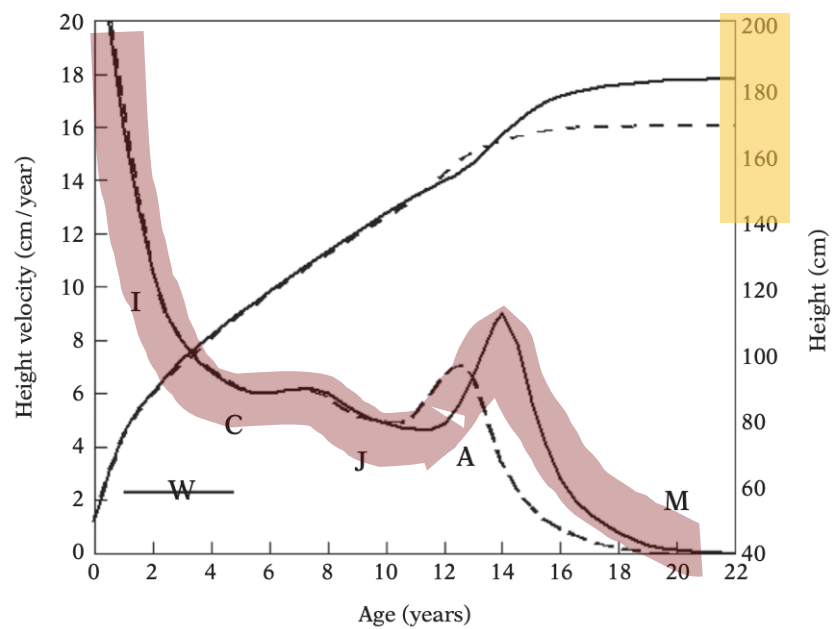
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**Goal 1: Understand
biological regulation
of growth in different
life history phases**

Today's framework

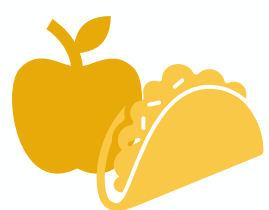
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Hormonal pathways control growth



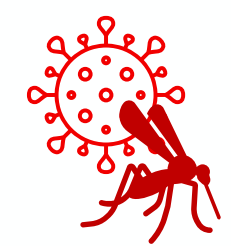
And these reflect long-term genetic adaptation (in many cases preceding the hominin lineage)



Nutrition

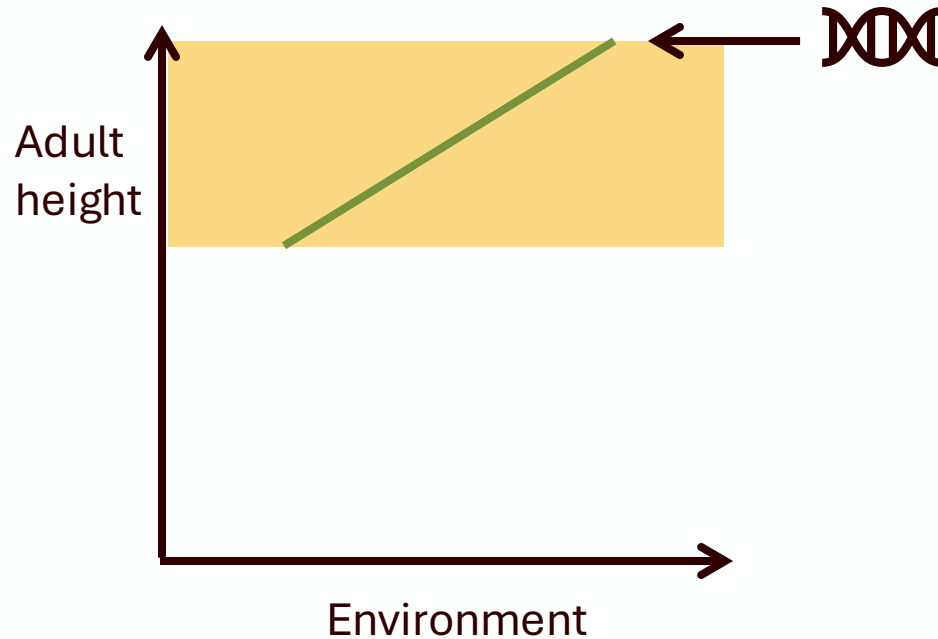
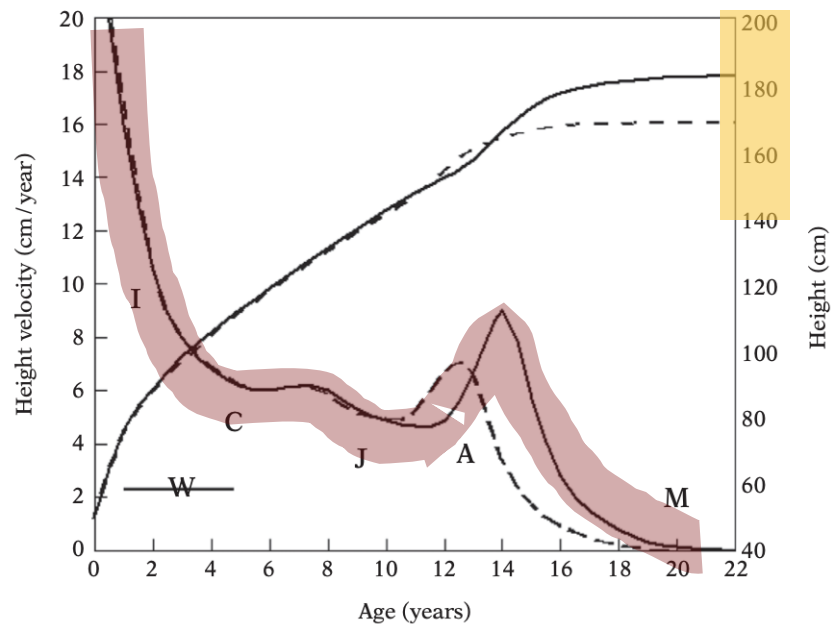


Sanitation, housing, wealth

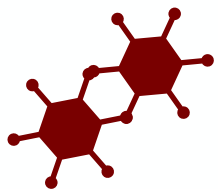


Disease

Today's framework



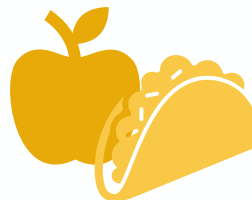
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Hormonal pathways control growth



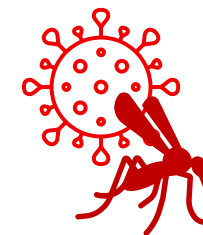
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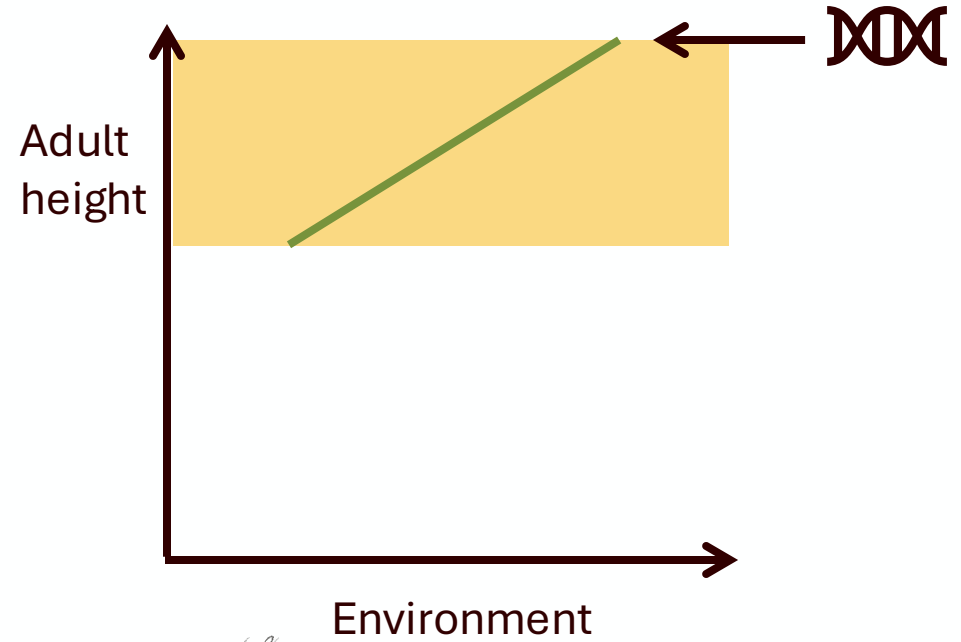
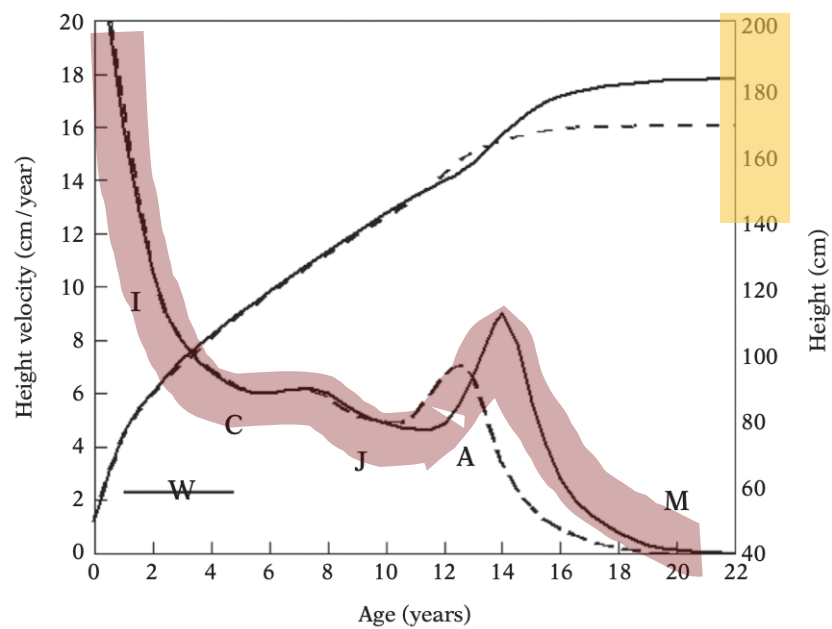


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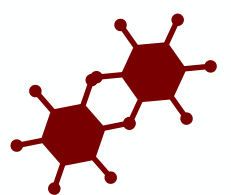
Disease

Today's framework



Goal 1: Understand biological regulation of growth in different life history phases

Goal 2: Compare importance of biocultural factors across human development

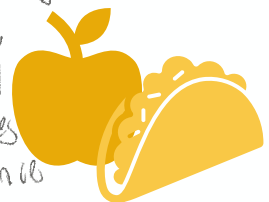


Hormonal pathways control growth



And these reflect long-term genetic adaptation (in many cases preceding the hominin lineage)

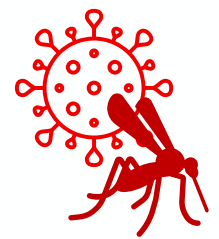
Differences here produce normal differences that influence rates of growth



Nutrition

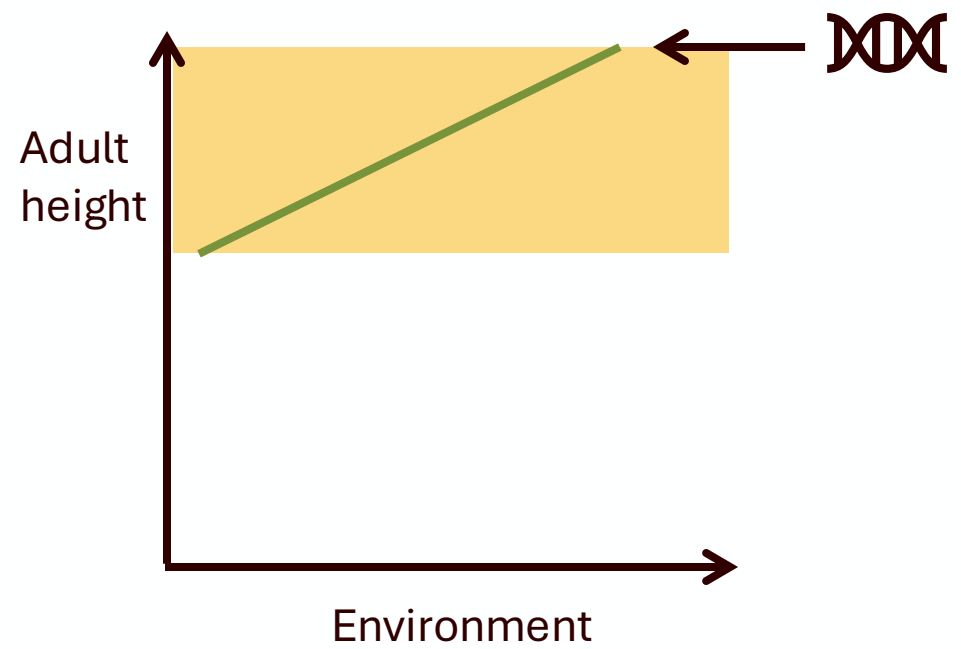
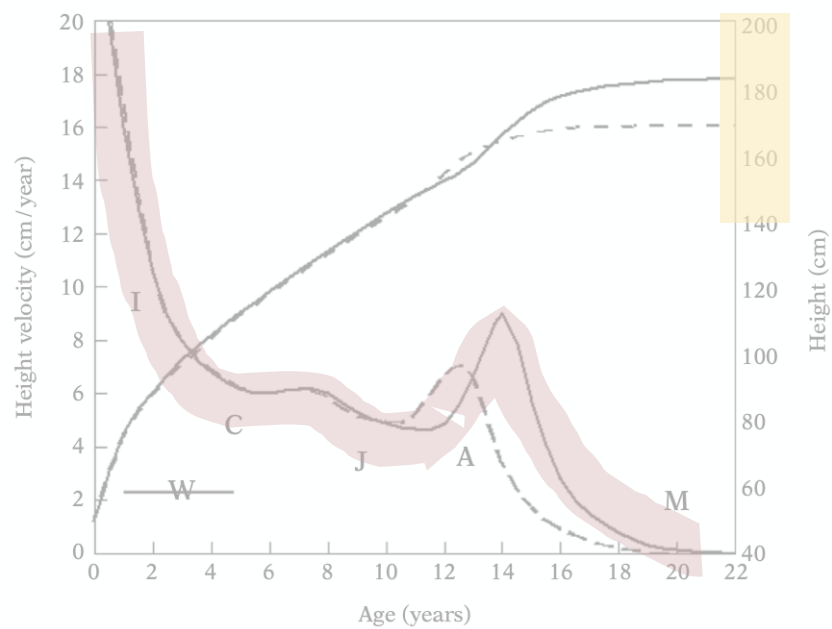


Sanitation, housing, wealth



Disease

Today's framework



Goal 1: Understand biological regulation of growth in different life history phases

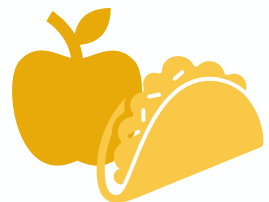
Goal 2: Compare importance of biocultural factors across human development



Hormonal pathways control growth



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Nutrition

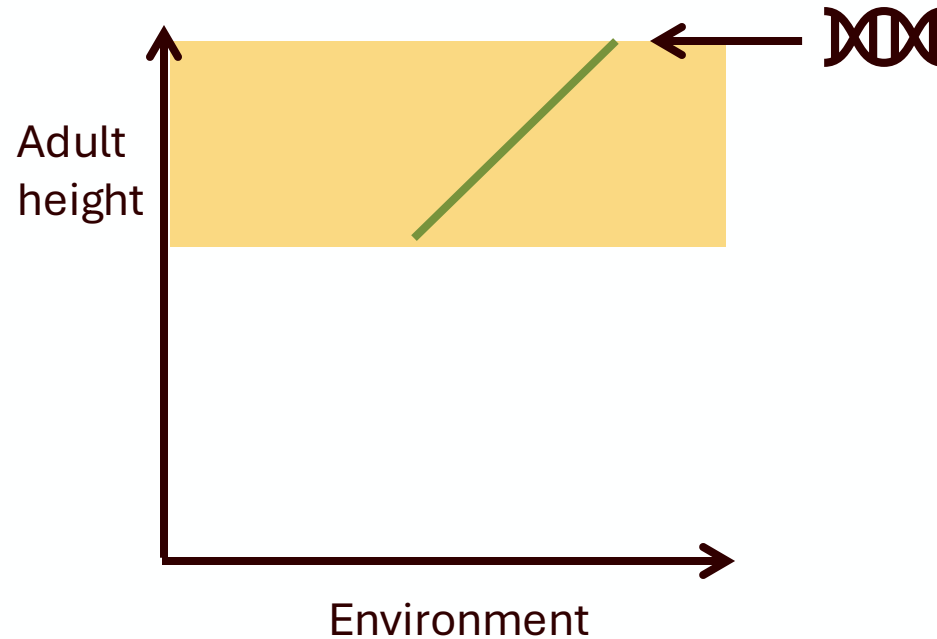
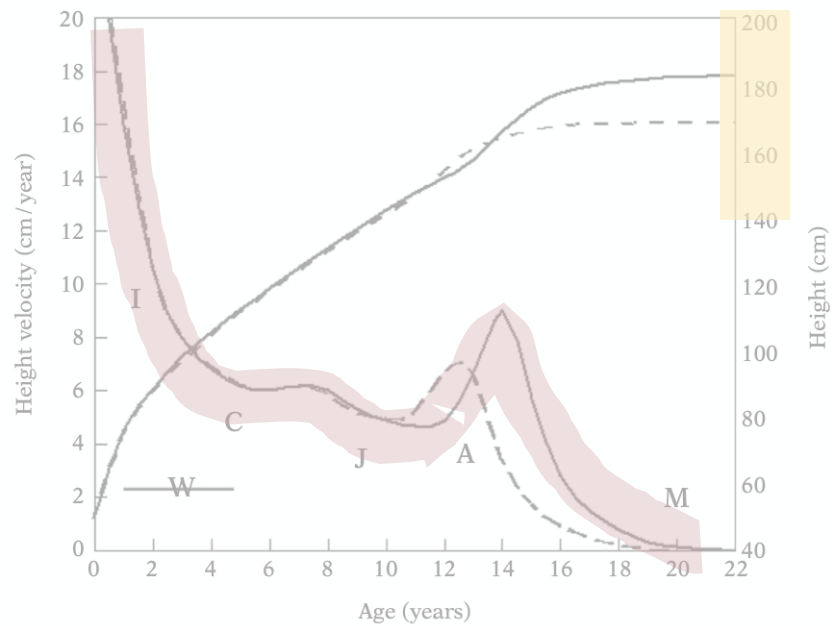


Sanitation, housing, wealth



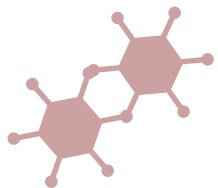
Disease

Today's framework



Goal 1: Understand biological regulation of growth in different life history phases

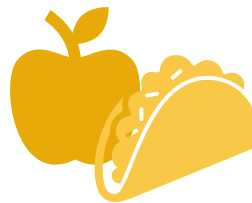
Goal 2: Compare importance of biocultural factors across human development



Hormonal pathways control growth



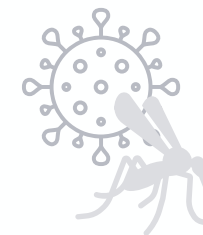
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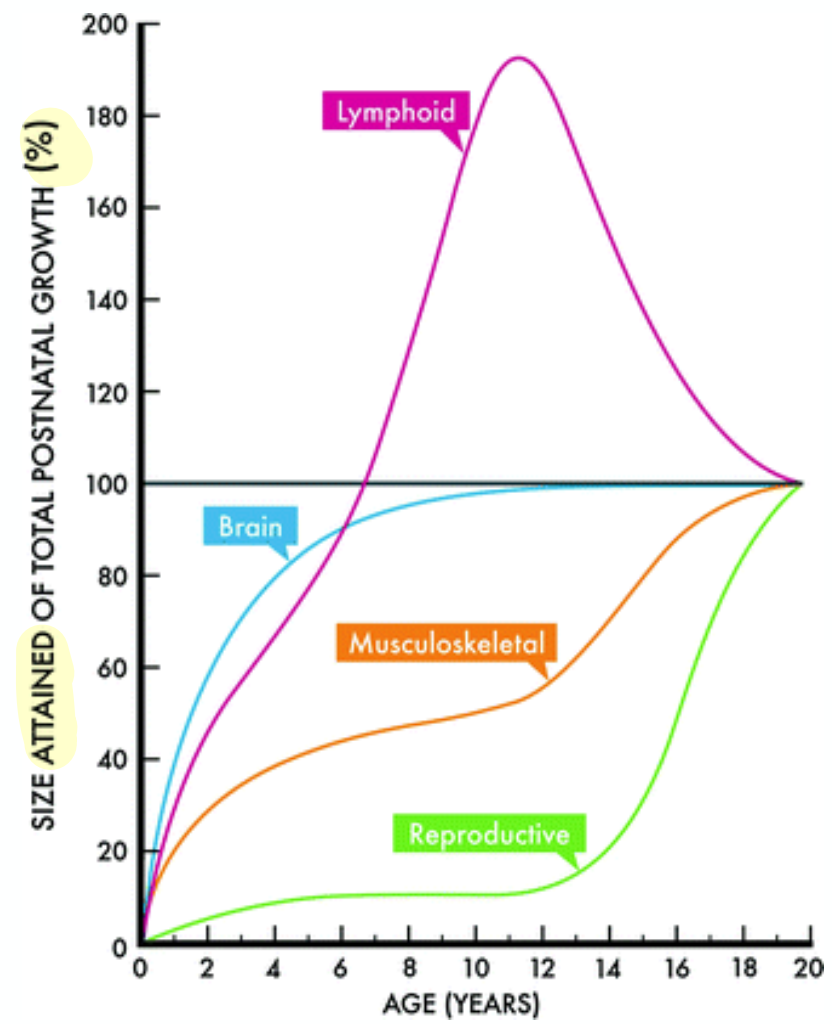


Sanitation, housing, wealth

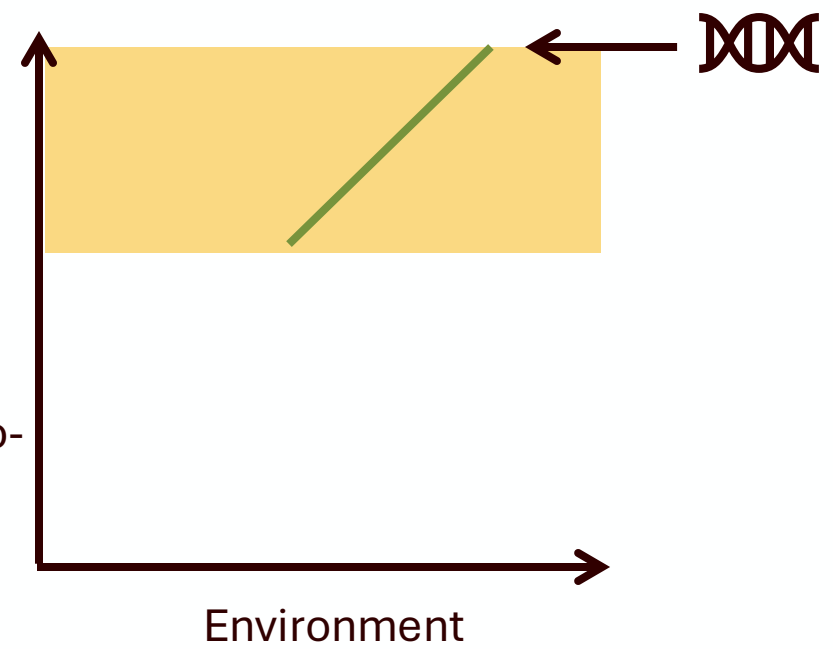


Disease

Today's framework



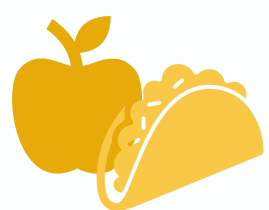
Adult brain size
Adult weight
Adult musculo-skeletal system



Goal 1: Understand biological regulation of growth in different life history phases

Goal 2a: Compare importance of biocultural factors across human development

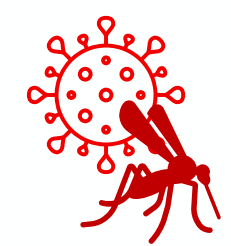
Goal 2b: ...and between traits



Nutrition

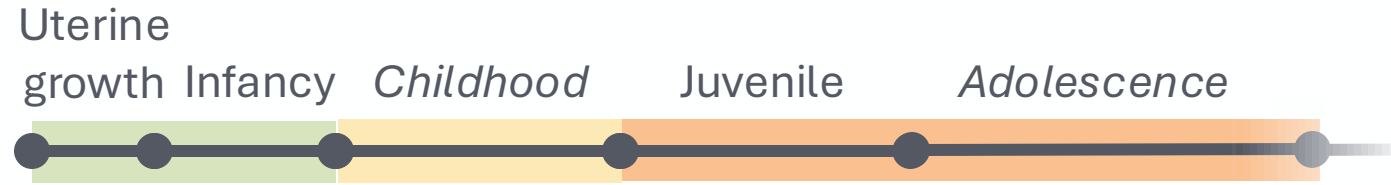


Sanitation, housing, wealth



Disease

↳ tradeoff to think of - brain growth vs body growth



Most research

Least research

Wot influence
of environmental
factors on growth

Pre- and perinatal growth

Uterine

growth

Infancy

Childhood

Juvenile

Adolescence



Evolution

Framework

Pre-, perinatal

Childhood & juv.

Adol.

Pre- and perinatal growth

Uterine

growth

Infancy Childhood Juvenile Adolescence



First trimester

Fertilization to 12th week: embryogenesis

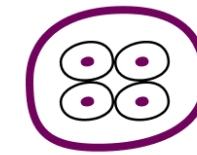
Second trimester

Fourth through sixth lunar month: rapid growth in length

Third trimester

Seventh lunar month to birth: rapid growth in weight and organ maturation

- Highest growth velocity
- Tightly genetically controlled, but...
- Highly sensitive to the environment



Zygote



Embryo

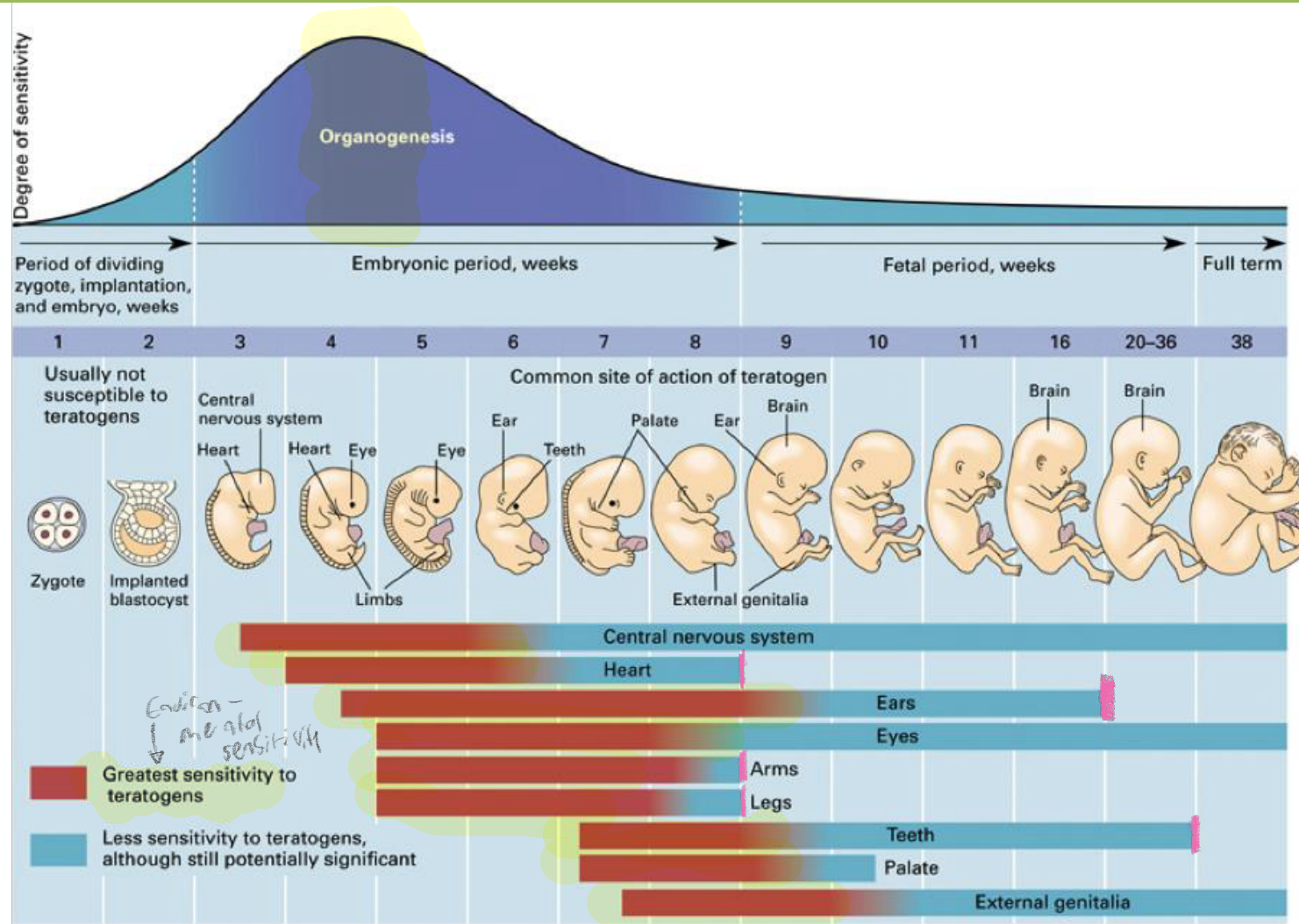


Fetus



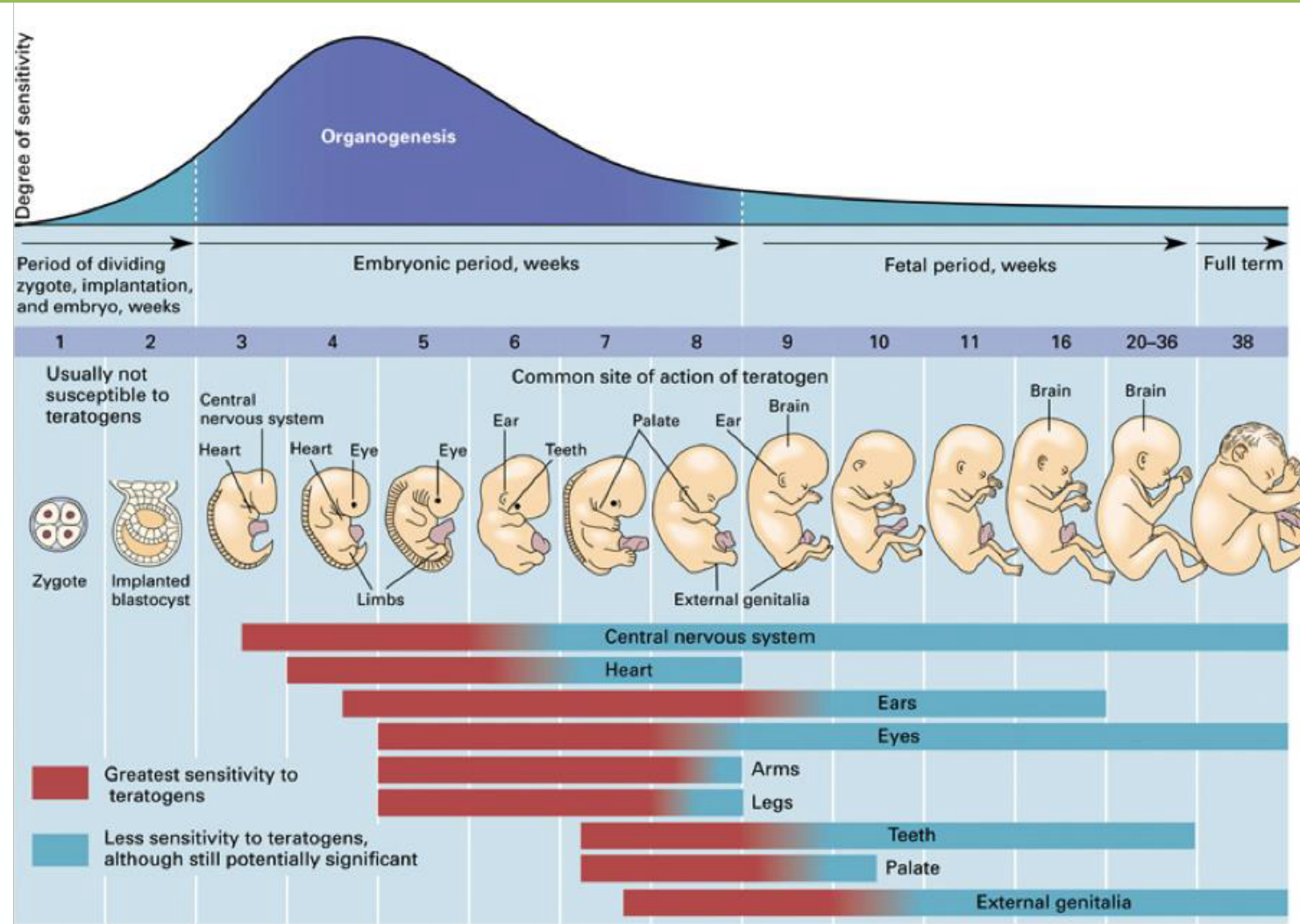
Baby

only after birth



Stopping point -
not sensitive to
environment at all anymore

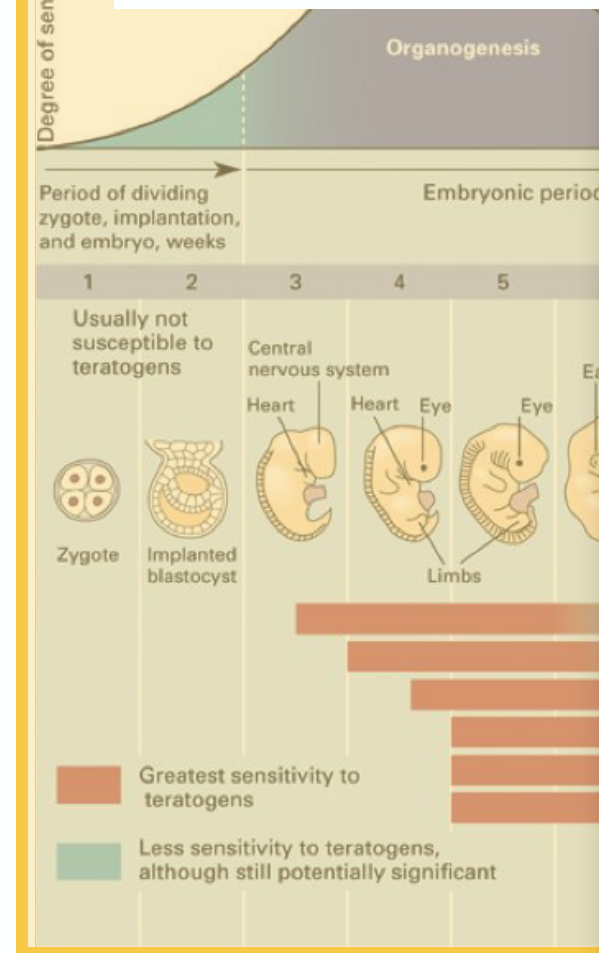
- Highest growth velocity
- Tightly genetically controlled, but...
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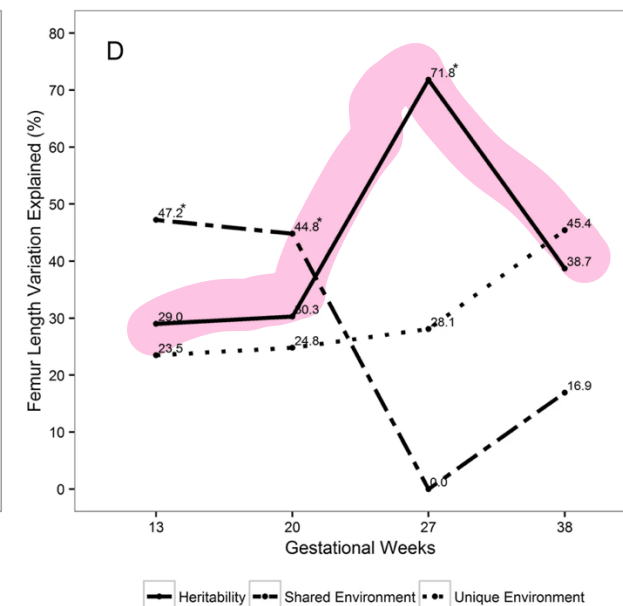
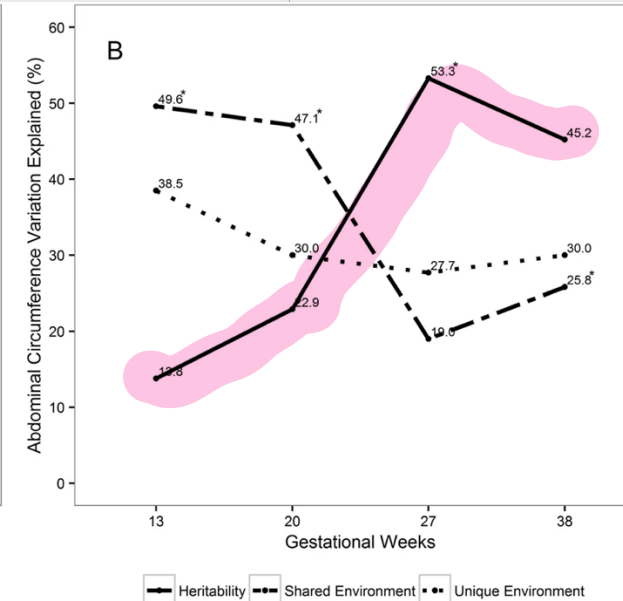
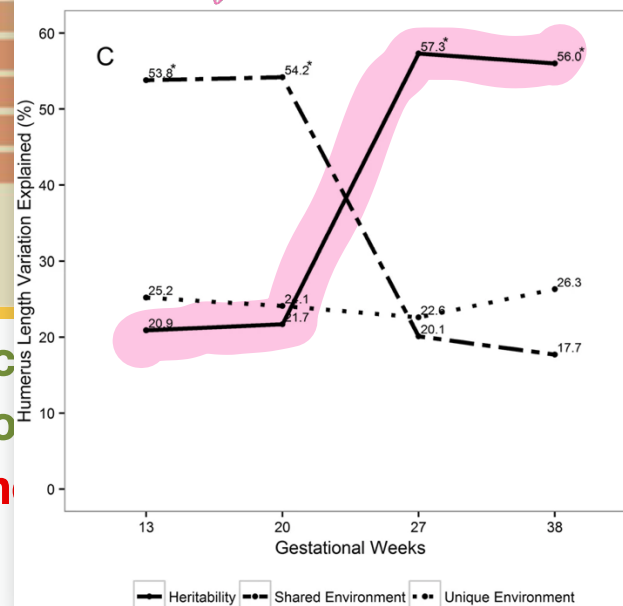
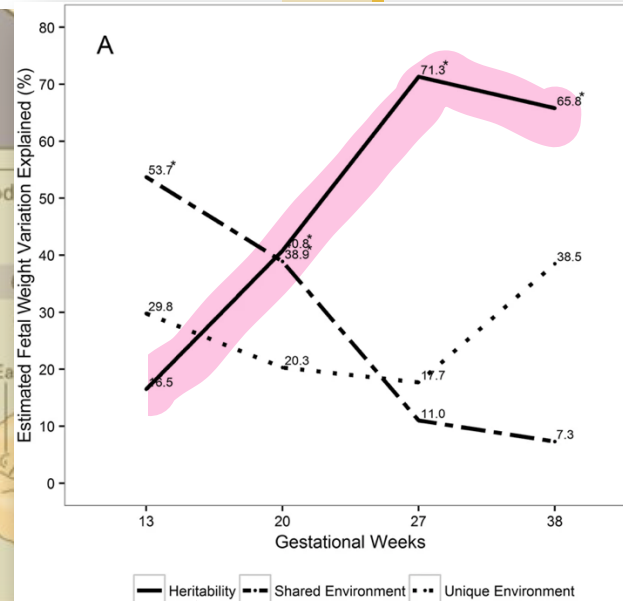
- **Highest growth velocity**
- **Tightly genetically controlled, but...**
- **Highly sensitive to the environment**

- **Complexity:** Many processes happening concurrently: one factor can have huge impact *and* this impact is increased by the interrelatedness of the processes
- **Rate:** Highest growth velocity: even brief perturbations can have large effects
- **Size:** Embryo & fetus small: diffusion of teratogens is quicker

1st trimester: highest sensitivity



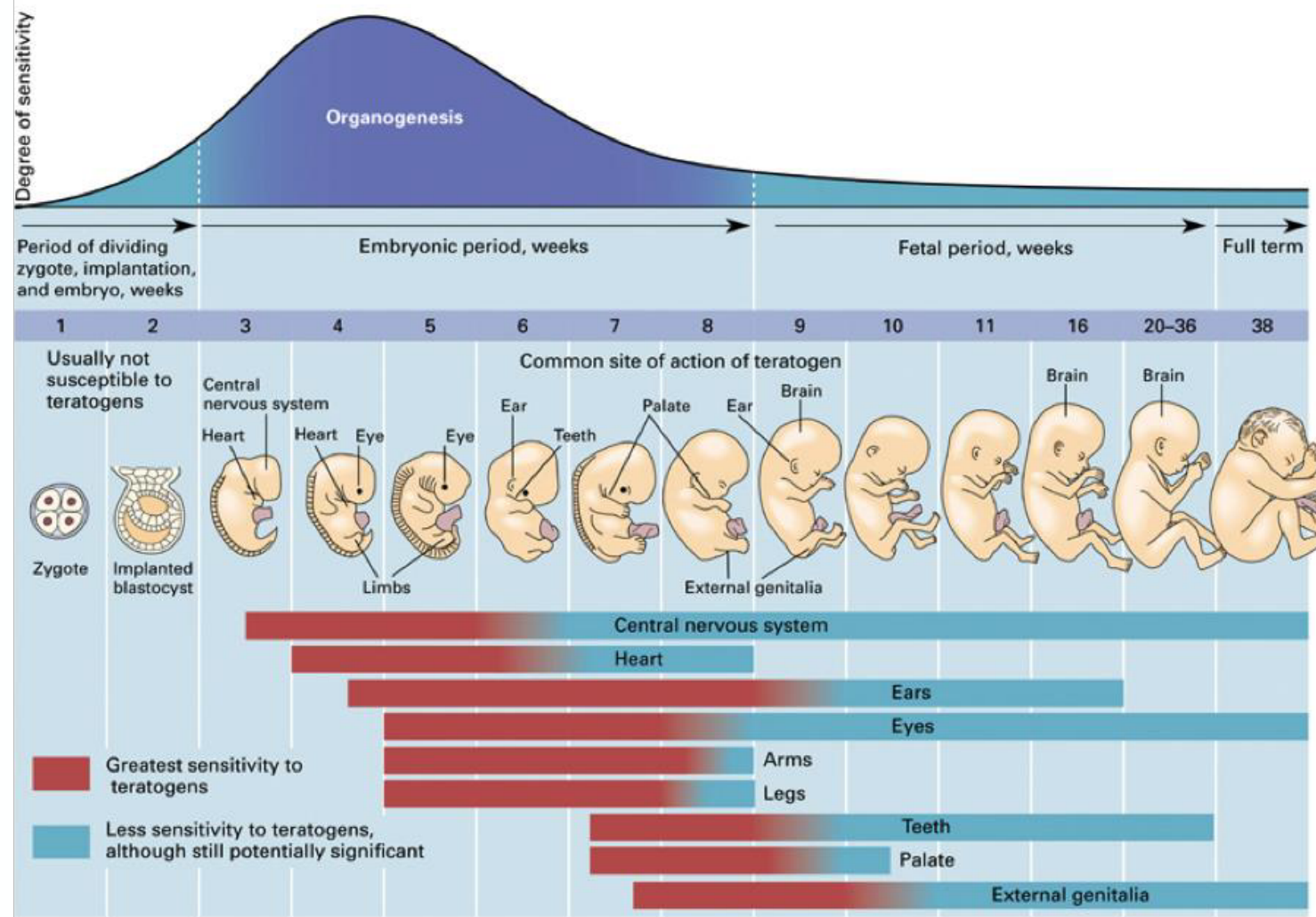
- Highest growth velocity
- Tightly genetically controlled
- Highly sensitive to the environment



Many processes occur concurrently: one factor can impact *and* this impact is the interrelatedness of the

fast growth velocity: even brief exposures can have large effects

mother & fetus small: diffusion of substances is quicker



- **Highest growth velocity**
- **Tightly genetically controlled, but...**
- **Highly sensitive to the environment**

- **Complexity:** Many processes happening concurrently: one factor can have huge impact *and* this impact is increased by the interrelatedness of the processes
- **Rate:** Highest growth velocity: even brief perturbations can have large effects
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Proximate drivers: not necessarily selected for

Pre- and perinatal growth

Uterine

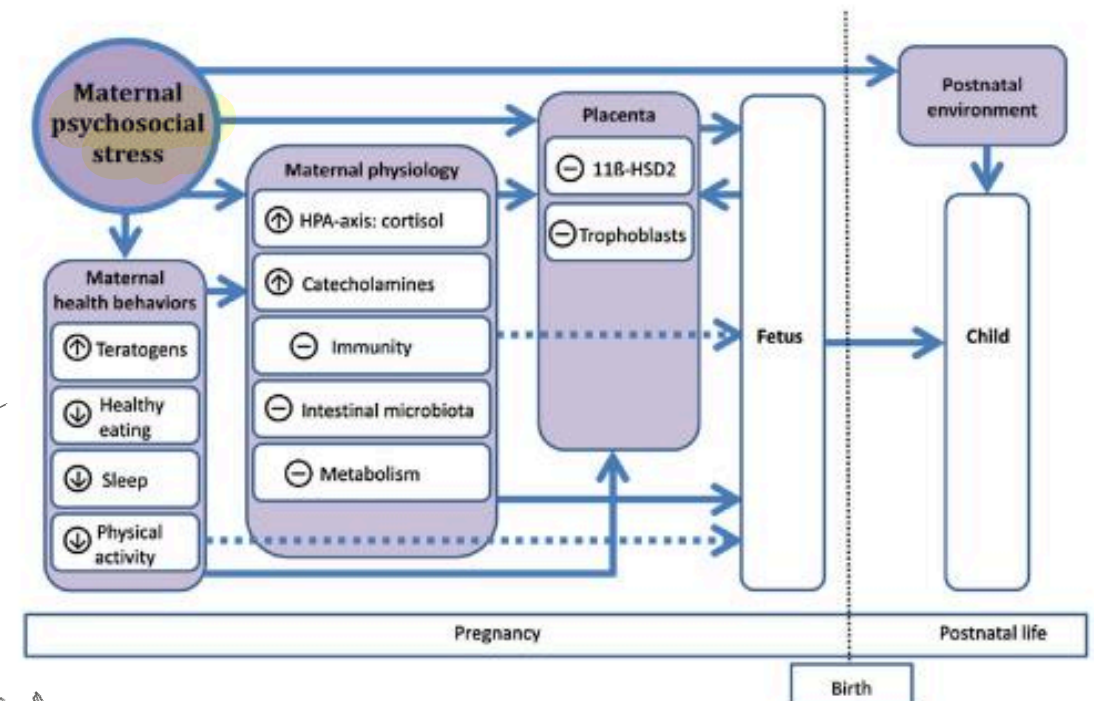
growth Infancy Childhood Juvenile



Pre- and post-natal **stress, anxiety, and depression** in mothers are associated with **impaired growth of offspring**

Fig. 1 An overview of the proposed mechanisms underlying maternal prenatal stress and child outcomes

Various pathways w/ how maternal psychological/social stress can influence growth of offspring



- Highest growth velocity
- Tightly genetically controlled, but...
- **Highly sensitive to the environment**

Maternal nutrition
Maternal stress

Pre- and perinatal growth

Uterine

growth **Infancy** *Childhood* Juvenile *Adolescence*



Evolution

Framework

Pre-, perinatal

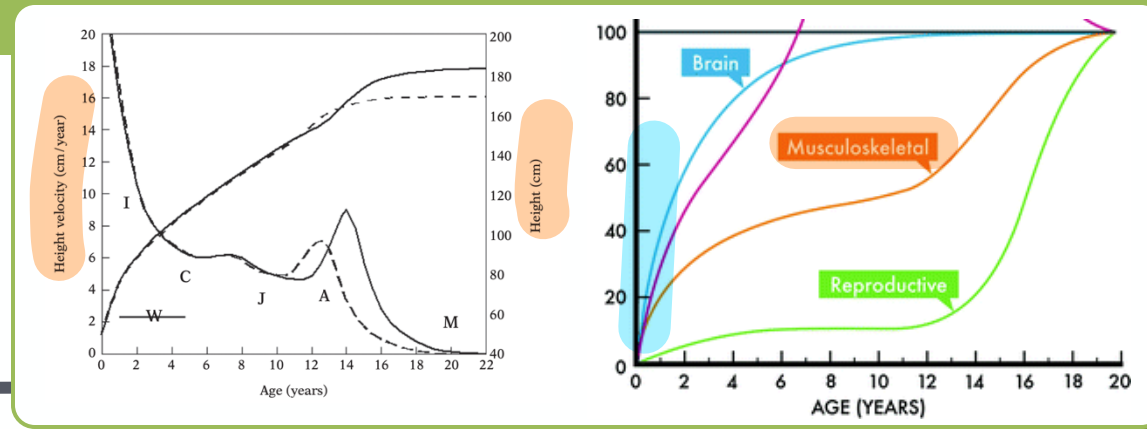
Childhood & juv.

Adol.

Pre- and perinatal growth

Uterine growth

Infancy Childhood Juvenile Adolescence



- **Rate:** High growth rate in brain and musculoskeletal system → higher rate of brain growth
- **Maternal nutrition:** Feeding exclusively by lactation to age 6 months and then lactation with gradual introduction of complimentary foods
Many developmental milestones in physiology, behaviour, and cognition

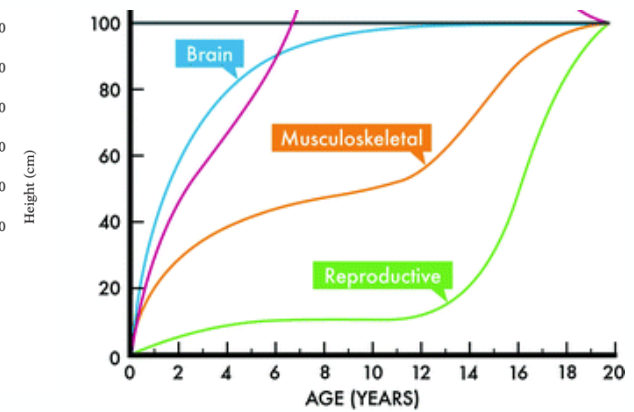
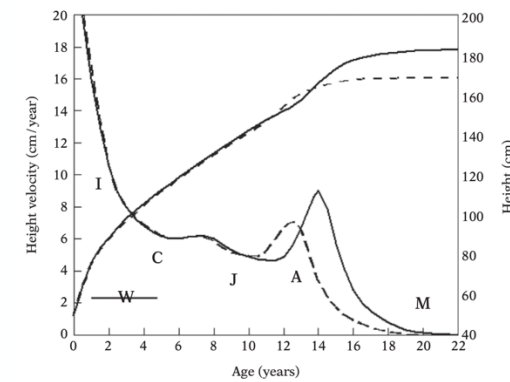
until 6 mos
can't eat
hard food
at all
bc teeth



Pre- and perinatal growth

Uterine

growth **Infancy** *Childhood* *Juvenile* *Adolescence*



- **Rate:** High growth rate in brain and musculoskeletal system
- **Maternal nutrition:** Feeding exclusively by lactation to age 6 months and then lactation with gradual introduction of complimentary foods

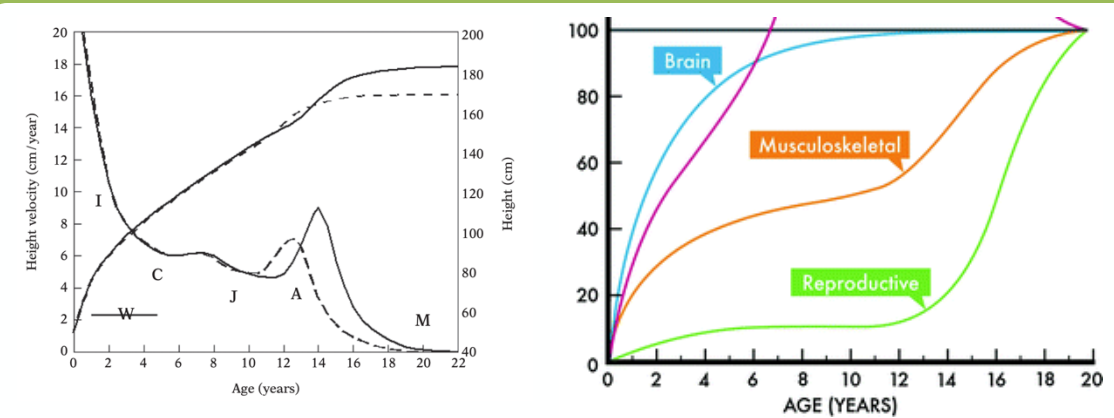


First 6 months:

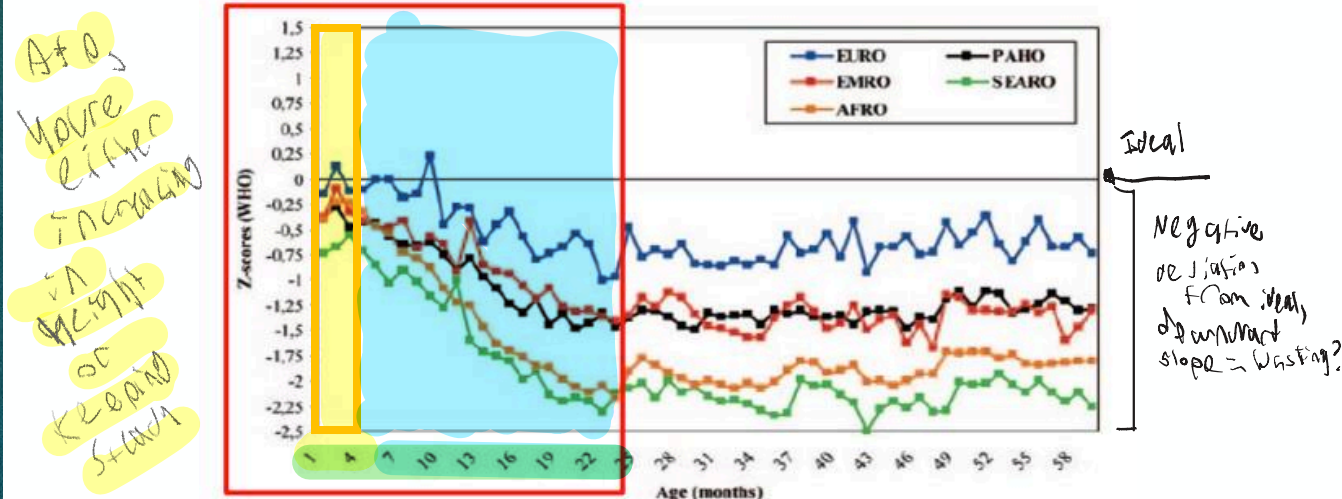
Relatively buffered from environment
Especially food-borne diseases

Pre- and perinatal growth

Uterine growth **Infancy** Childhood Juvenile Adolescence



- **Rate:** High growth rate in brain and musculoskeletal system
- **Maternal nutrition:** Feeding exclusively by lactation to age 6 months and then lactation with gradual introduction of complimentary foods

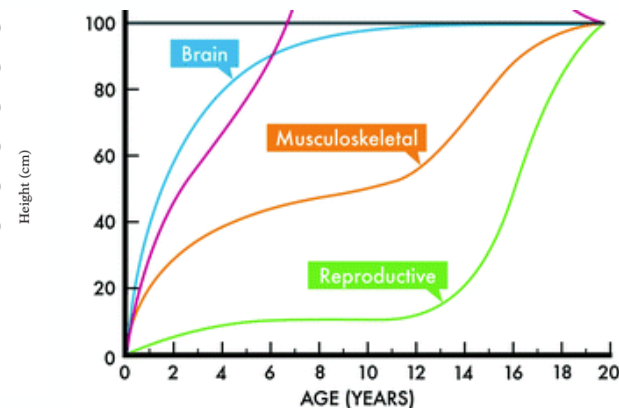
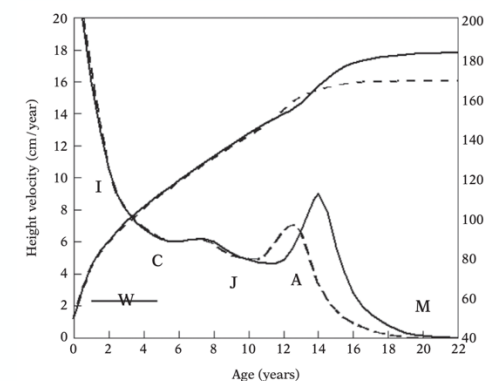


Victoria et al., 2010
Mean height-for-age z-scores by age, relative to the WHO growth standard, 54 countries.

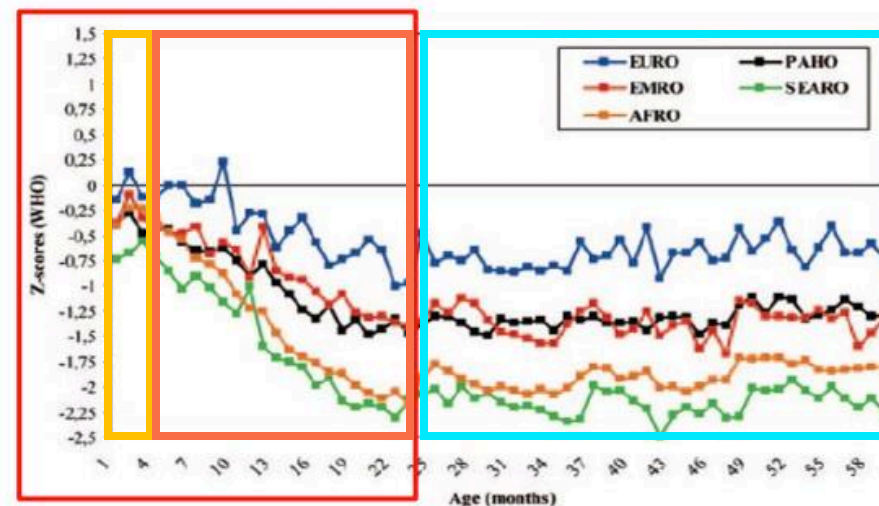
Pre- and perinatal growth

Uterine

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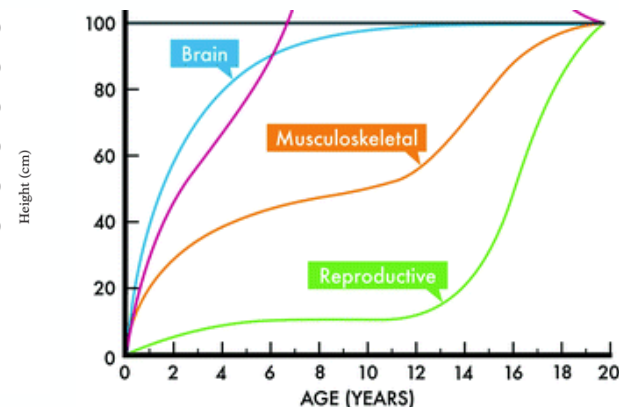
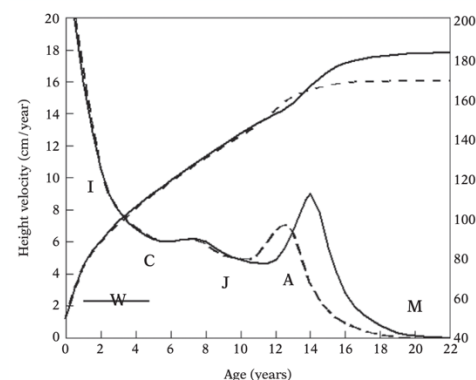


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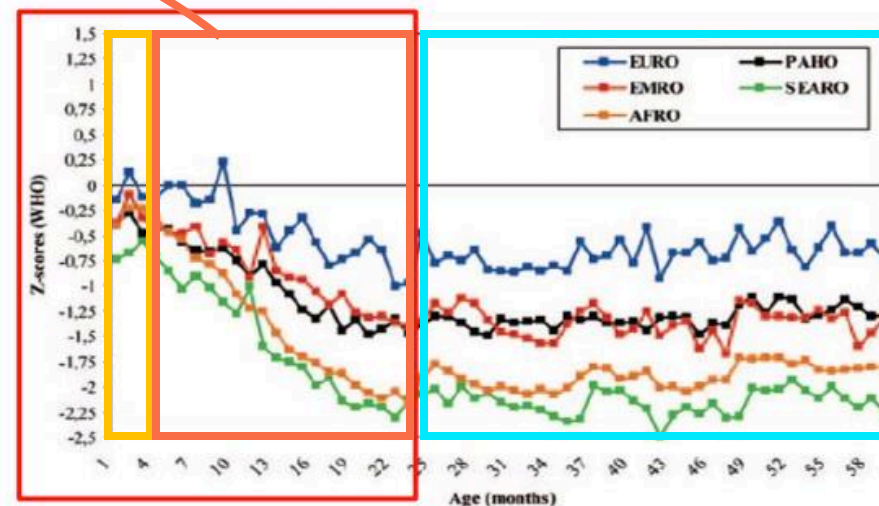
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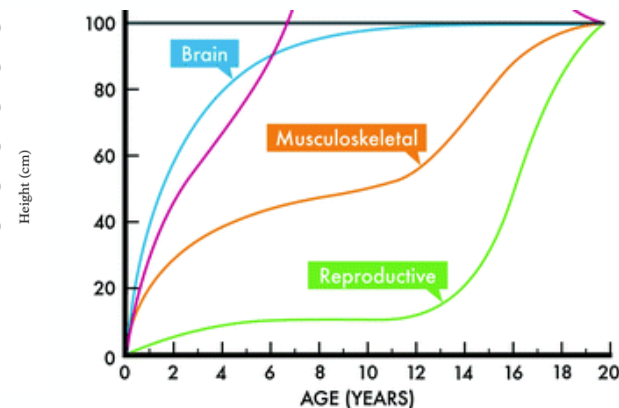
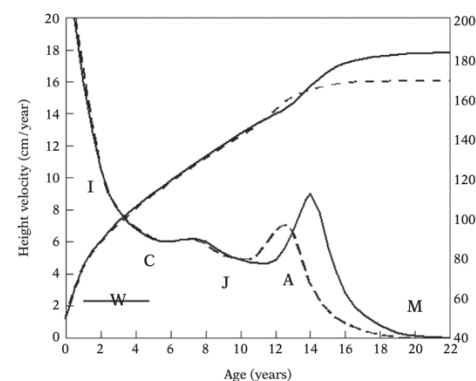
Growth faltering: tradeoffs



Victoria et al., 2010
Mean height-for-age z-scores by age, relative to the WHO growth standard, 54 countries.

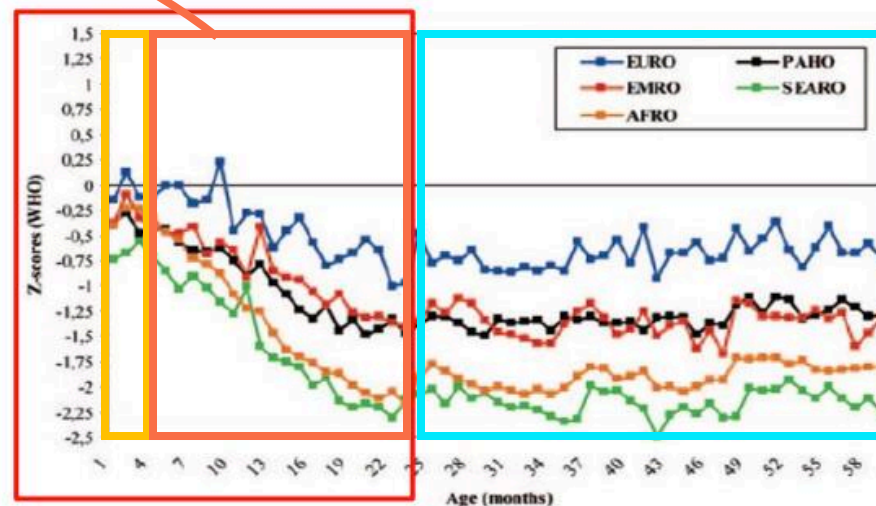
Pre- and perinatal growth

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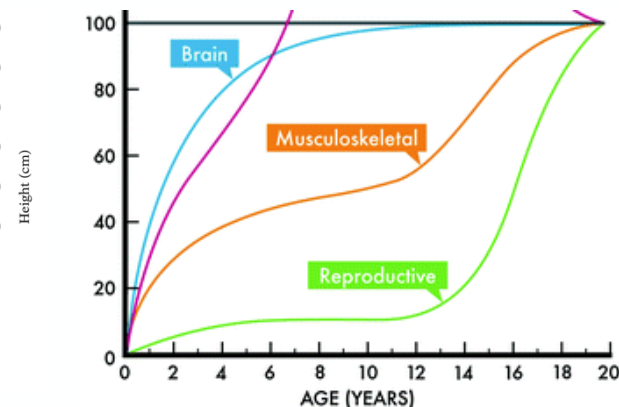
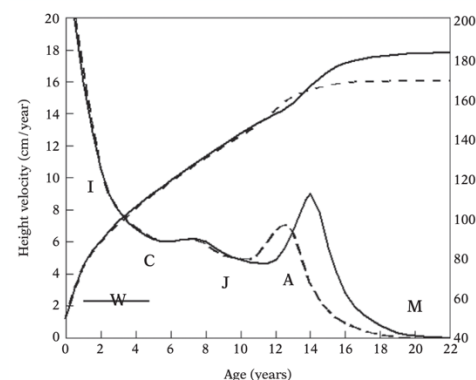
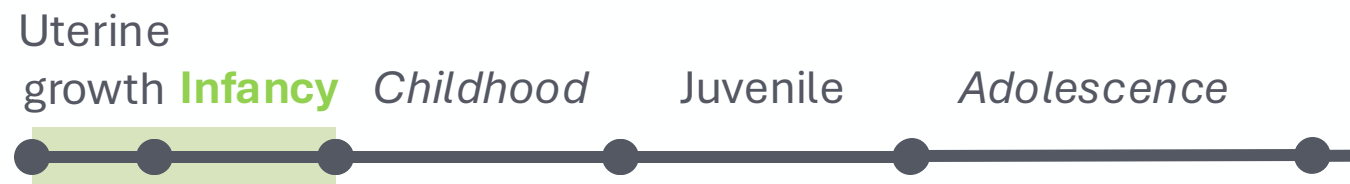


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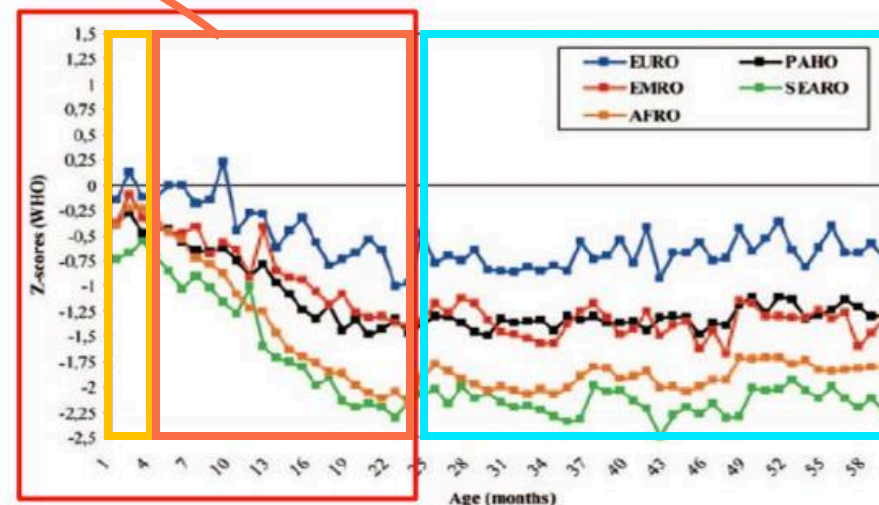
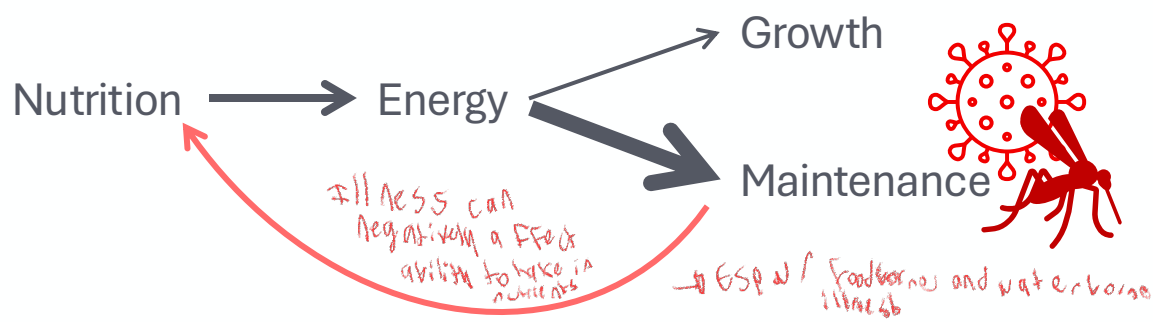
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Growth faltering: tradeoffs

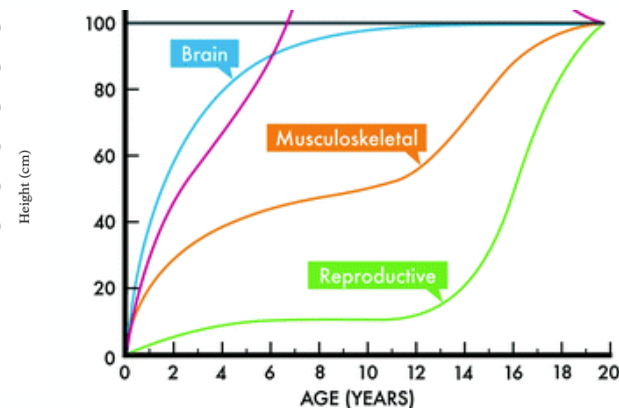
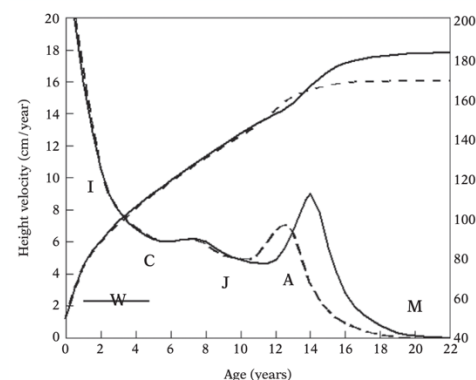
- Fundamentally, growth faltering results from competing demands between growth and maintenance (*immune function*)
- Complex interactions can exacerbate the impact of this tradeoff



Victoria et al., 2010
Mean height-for-age z-scores by age, relative to the WHO growth standard, 54 countries.

Pre- and perinatal growth

Uterine growth **Infancy** *Childhood* *Juvenile* *Adolescence*



for every child

Press centre
Donate

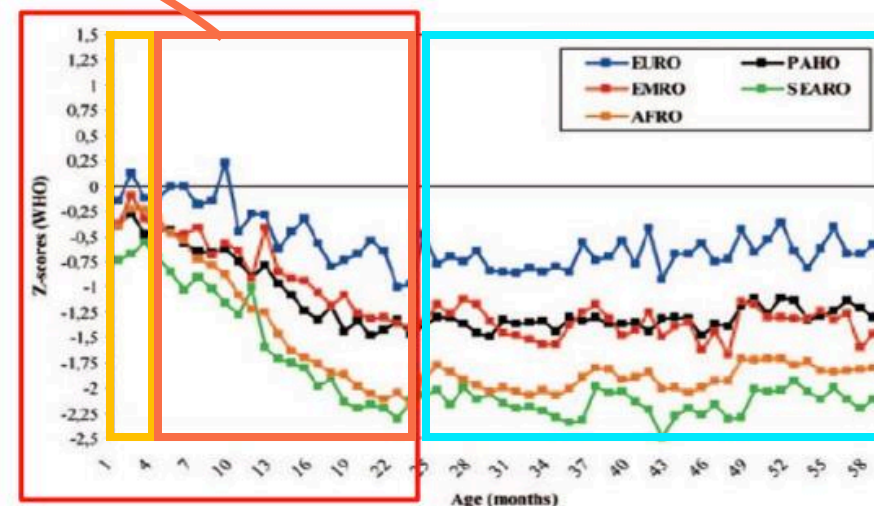
What we do
Research and reports
Stories
Who we are
Take action

Press release

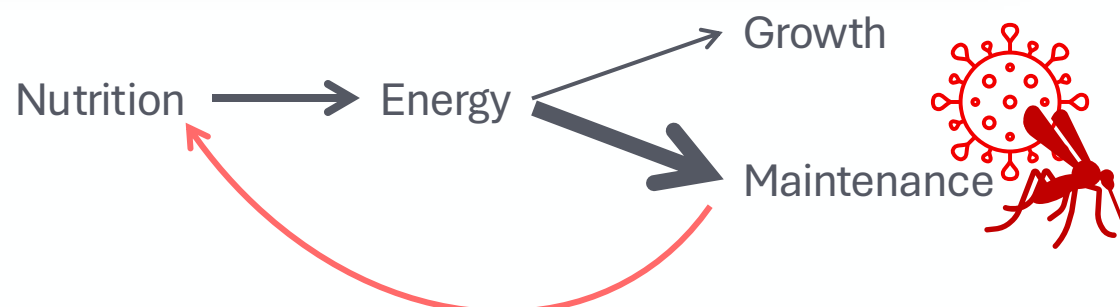
Children suffering dire drought across parts of Africa are 'one disease away from catastrophe' – warns UNICEF

Number of drought-hit people in Ethiopia, Kenya, Somalia without reliable access to safe water has risen from 9.5 million to 16.2 million in the space of five months, meanwhile children in the Sahel also facing extremely high levels of water vulnerability

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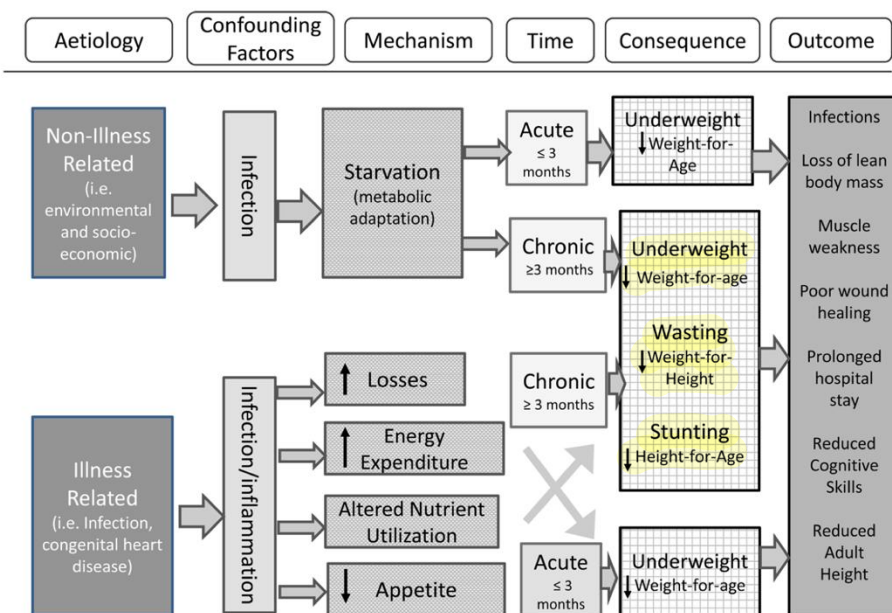
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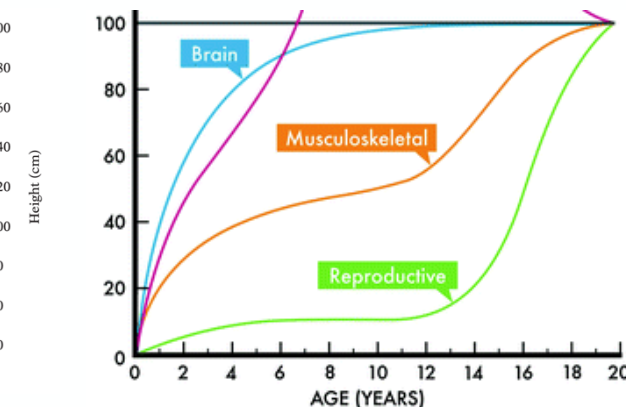
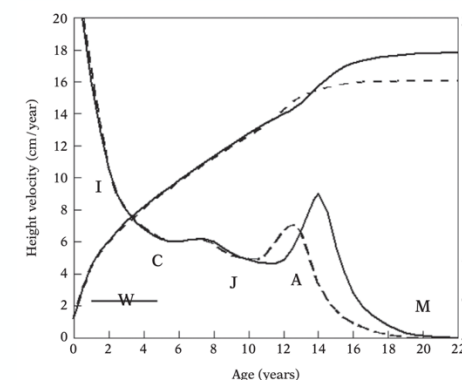
Pre- and perinatal growth

Catch-Up Growth in Infants and Young Children With Faltering Growth: Expert Opinion to Guide General Clinicians

*Richard Cooke, MD, †Olivier Goulet, MD, PhD, ‡Koen Huysentruyt, MD, PhD, §Koen Joosten, MD, PhD, ||Anuradha Vaman Khadilkar, MD, ¶Meng Mao, MD, ***Rosan Meyer, PhD, ††Andrew M. Prentice, PhD, and ‡‡Anul Singh, MD, MRCP



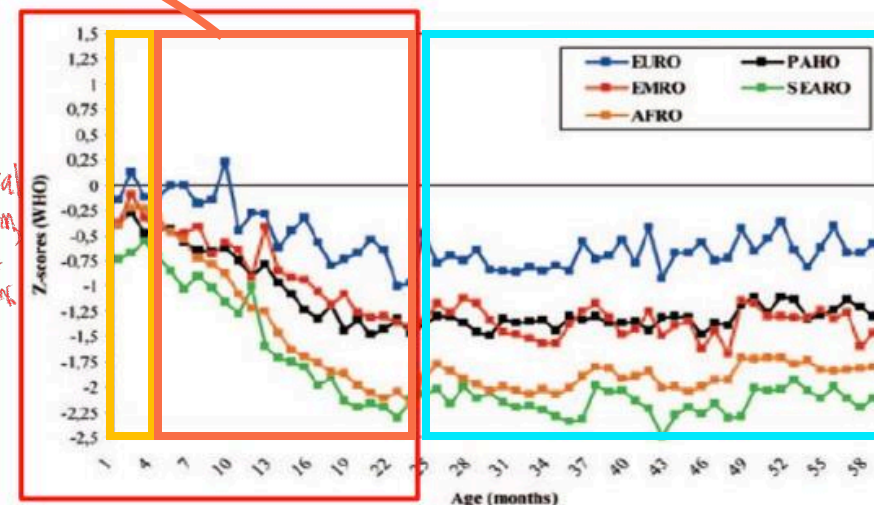
Adolescence



- **Rate:** High growth rate in brain and musculoskeletal system
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trying to determine cause of growth faltering

General term for wasting

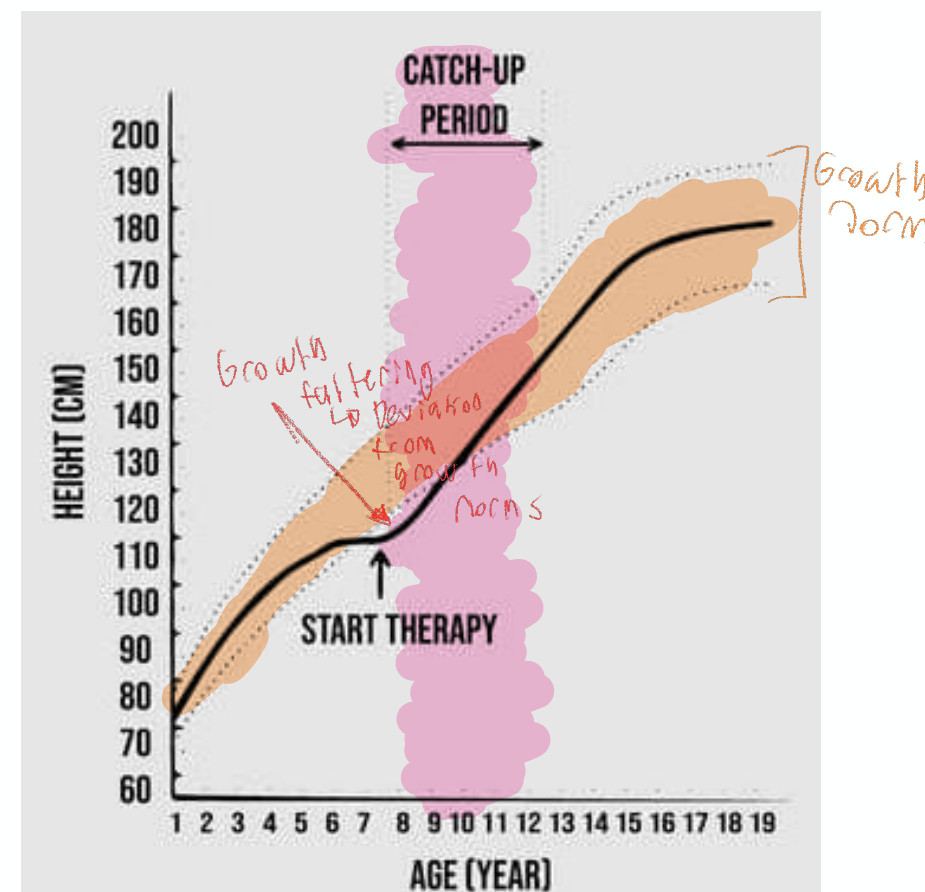
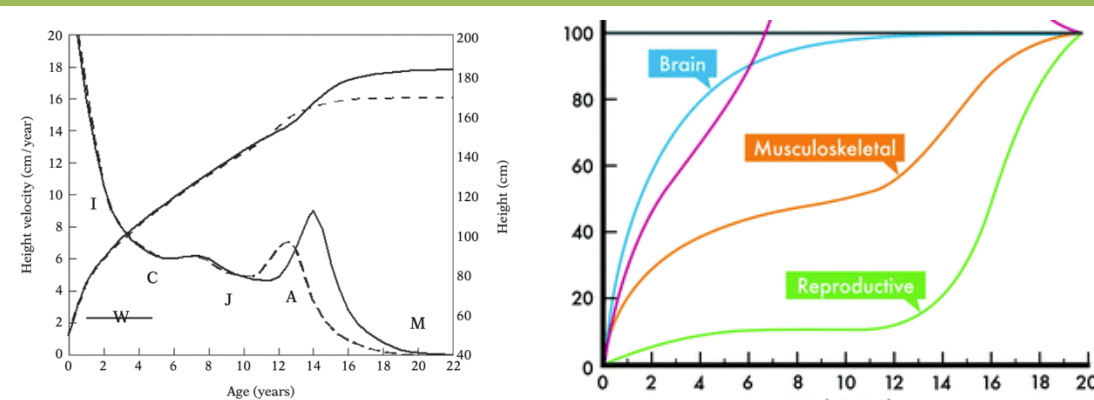
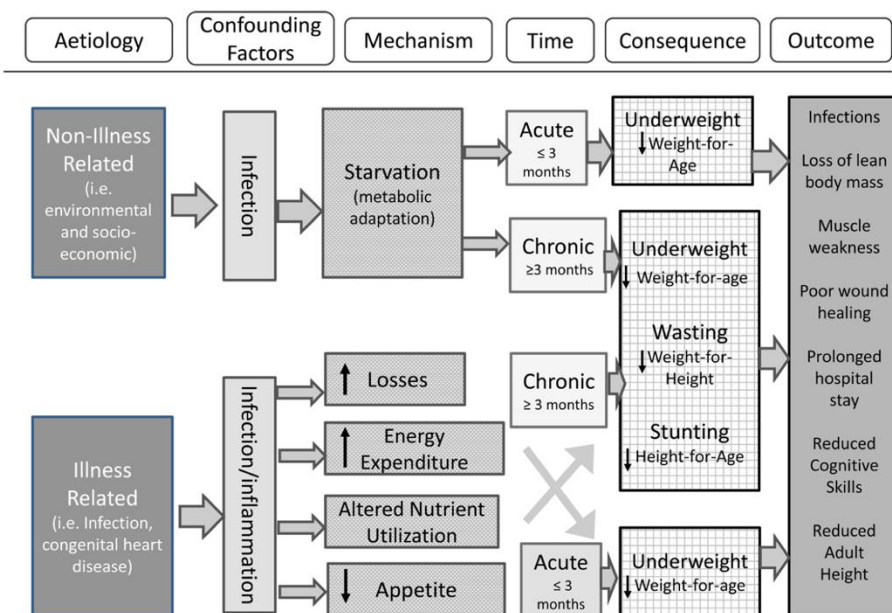


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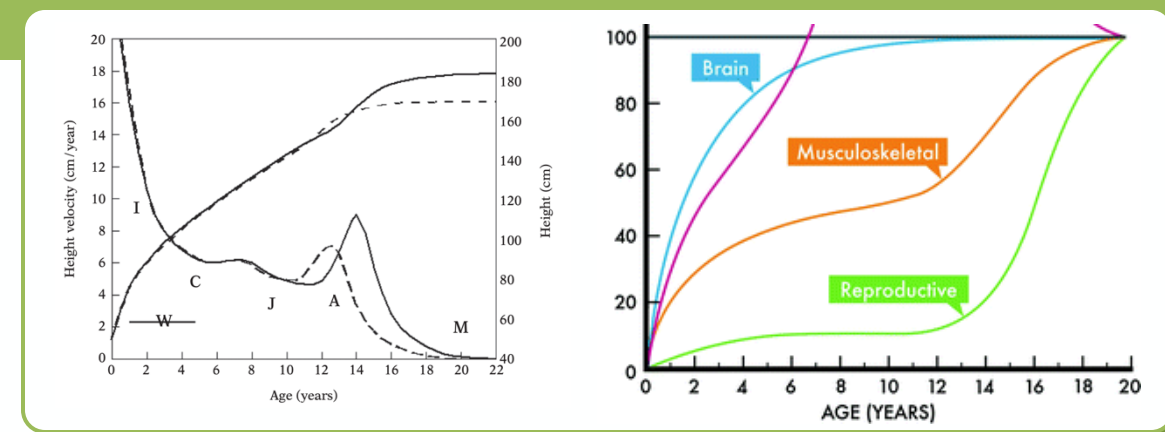
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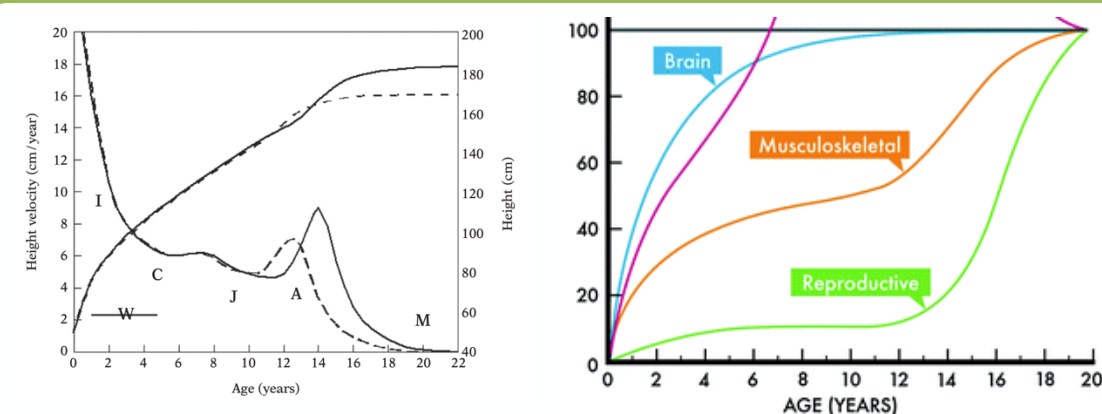
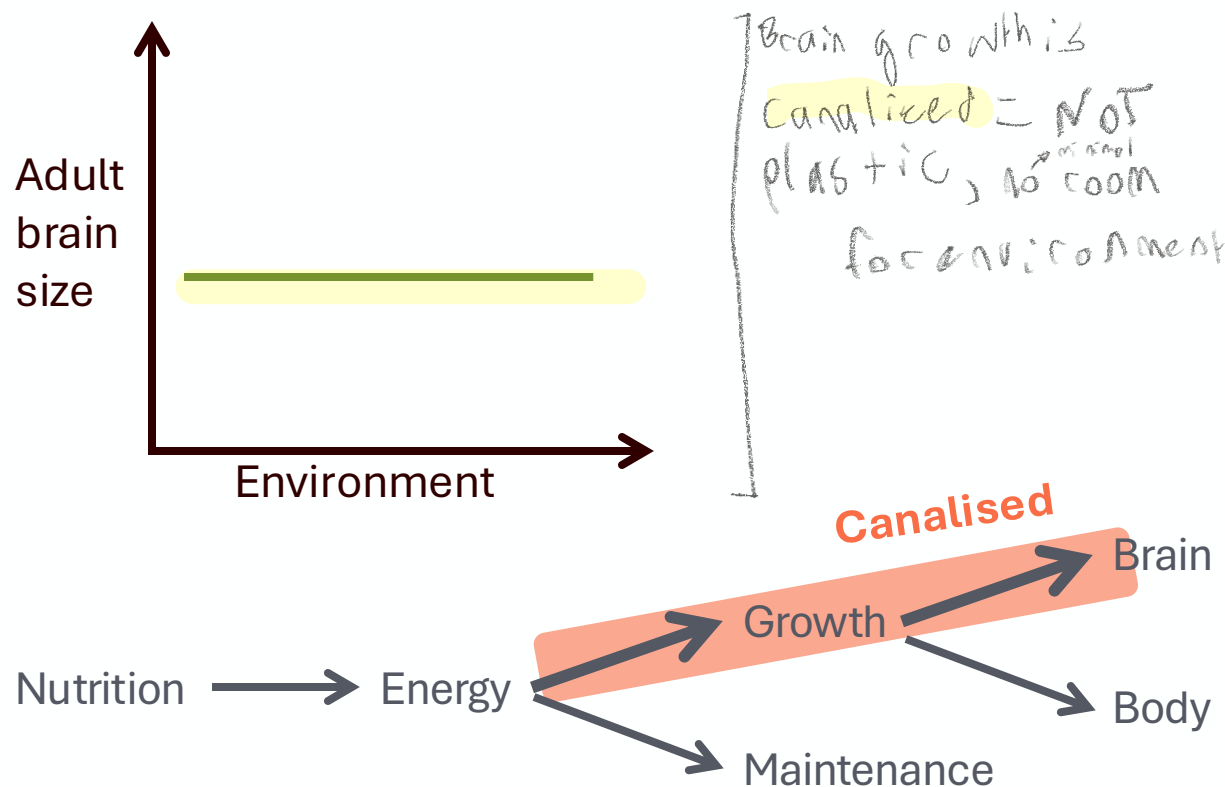
Pre- and perinatal growth



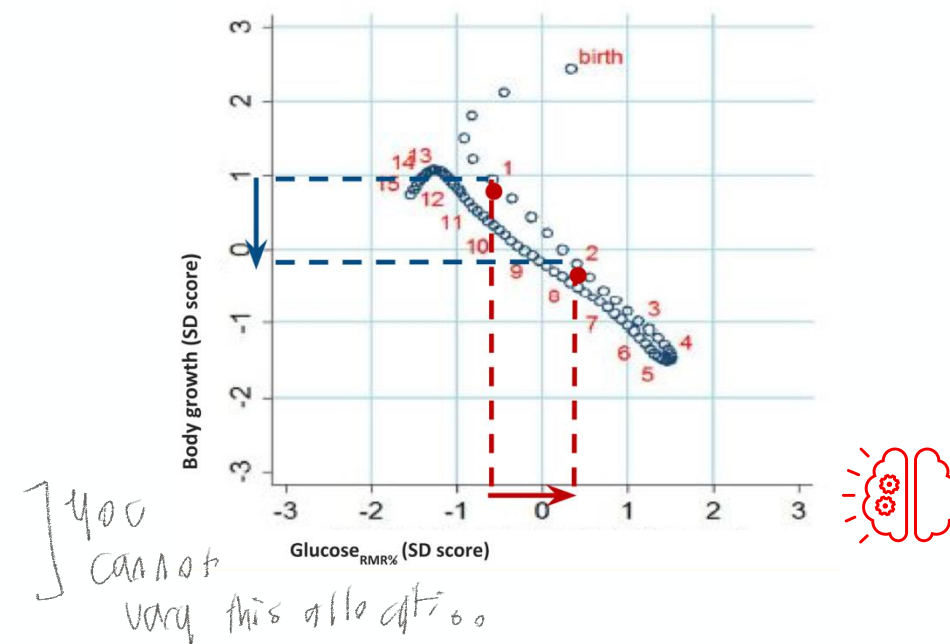
Pre- and perinatal growth



Brain growth is highly **canalized**...



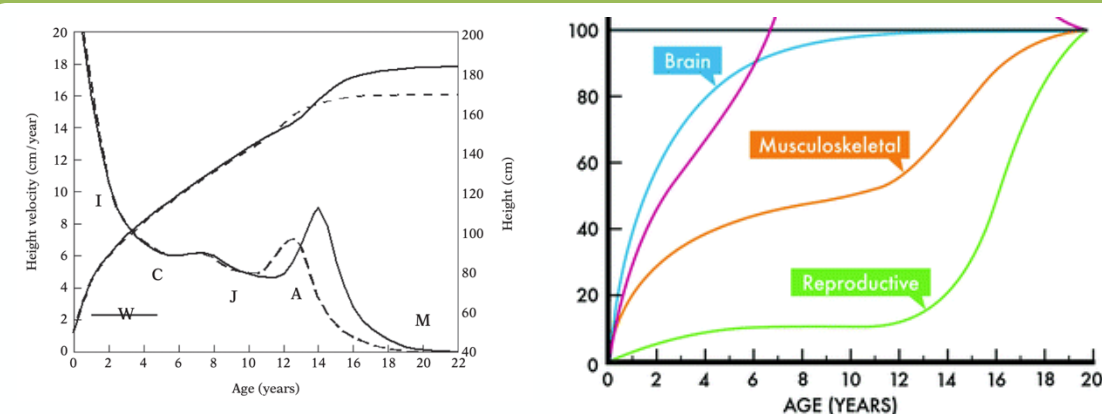
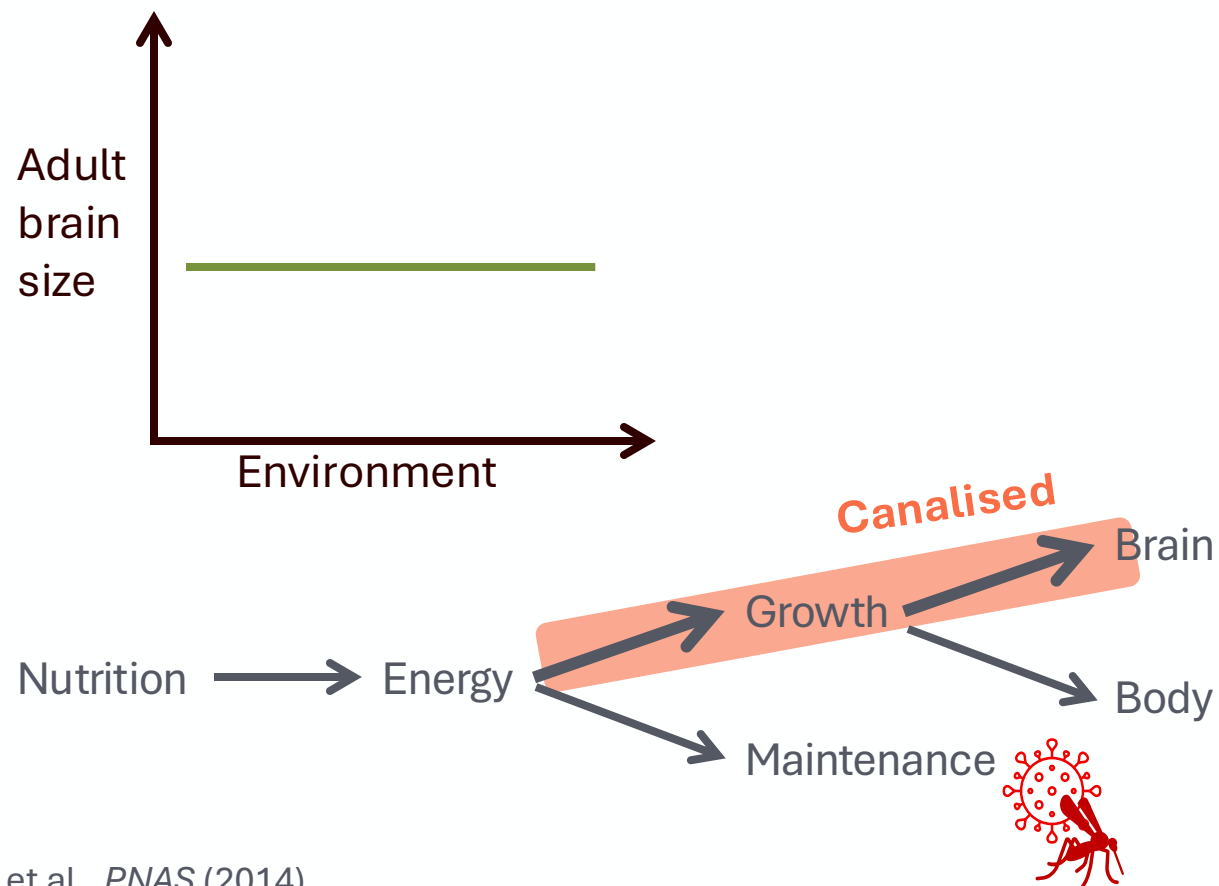
...and traded off with body growth



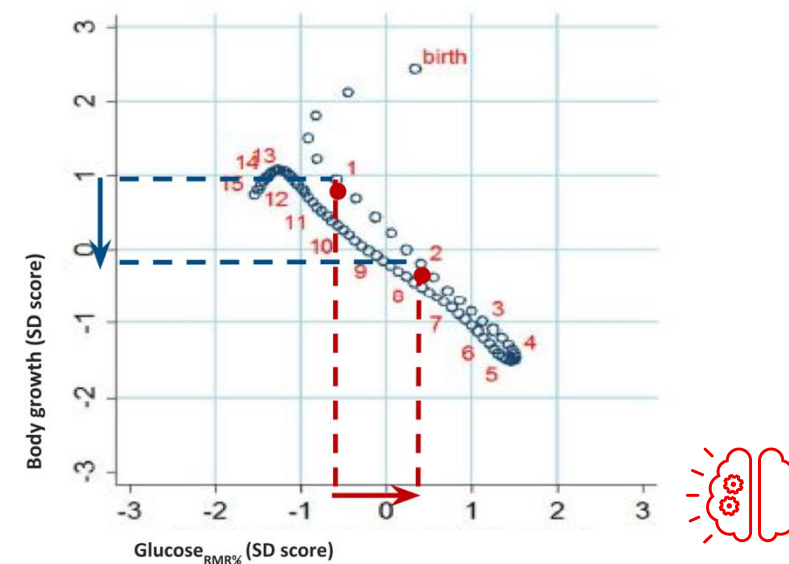
Pre- and perinatal growth



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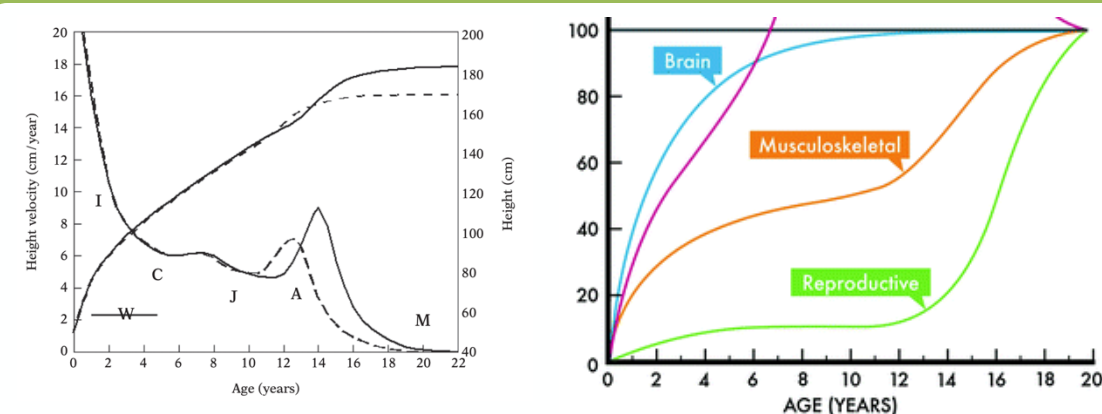
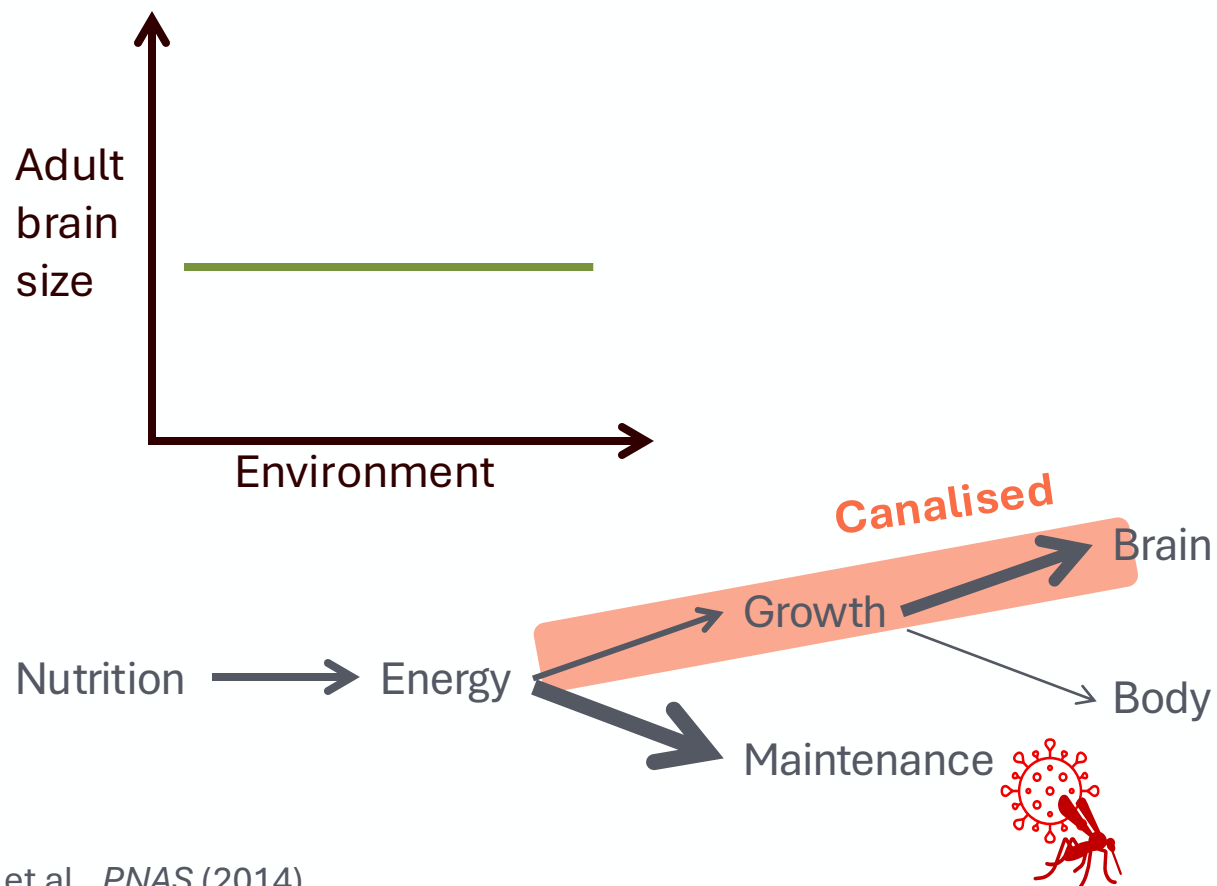
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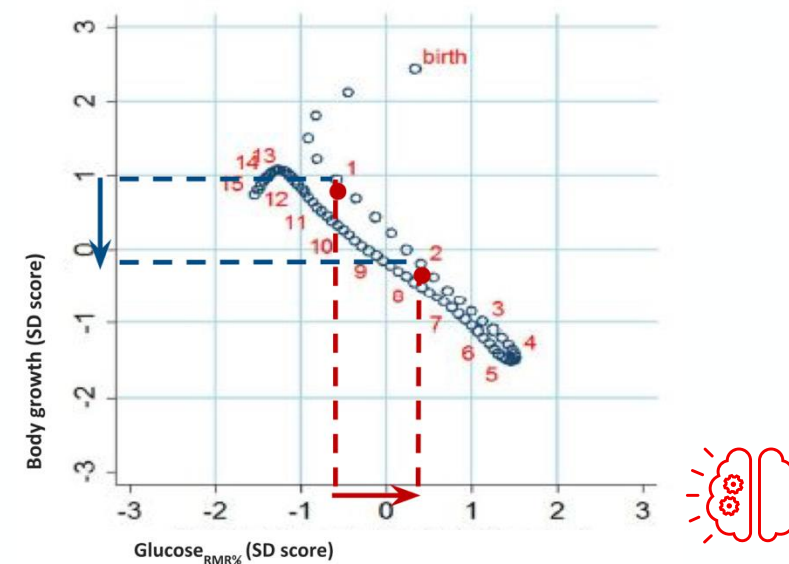
Pre- and perinatal growth



Brain growth is highly canalized...



...and traded off with body growth



Musculoskeletal growth is more sensitive to the environment than brain growth

Pre- and perinatal growth

Uterine

growth Infancy

Childhood

Juvenile

Adolescence



Body growth rates **highly** responsive to the environment *(though buffered for a bit: breastfeeding)*

- Big effects
- **But also: scope for interventions with big impact**

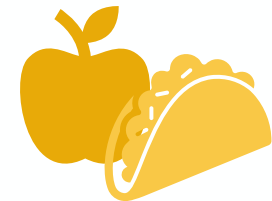
How?

Nutrition is key: provides energy. Infection & other drivers act by driving energy investment away from growth or affecting energy intake

- Complexity & rate
- Linear growth during fetal-neonatal-infancy periods: primarily driven by **insulin-like growth factors (IGFs)** and **insulin**

Hormones

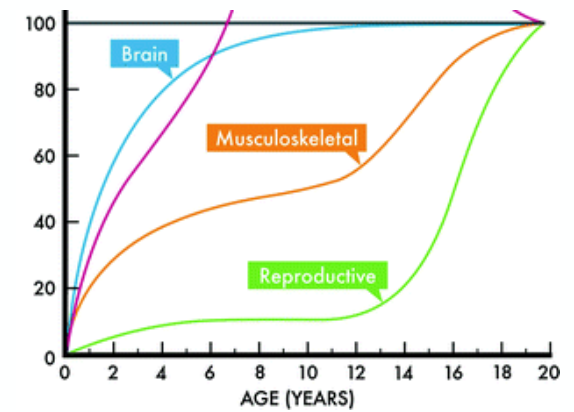
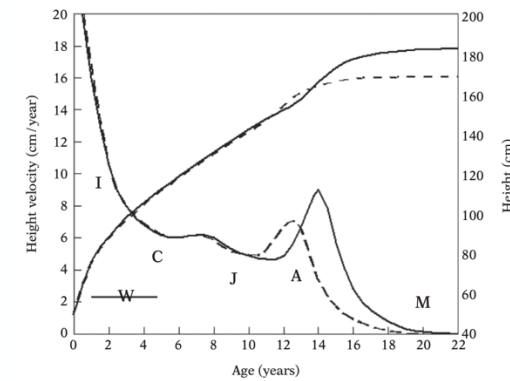
Proximate mechanism of glucose availability affecting insulin, affects growth



Childhood and juvenile growth

Uterine

growth Infancy *Childhood* Juvenile *Adolescence*



- Linear growth during childhood & juvenile periods: primarily driven by **insulin-like growth factors (IGFs)** and **insulin**

amnetwork

Pre-, perinatal

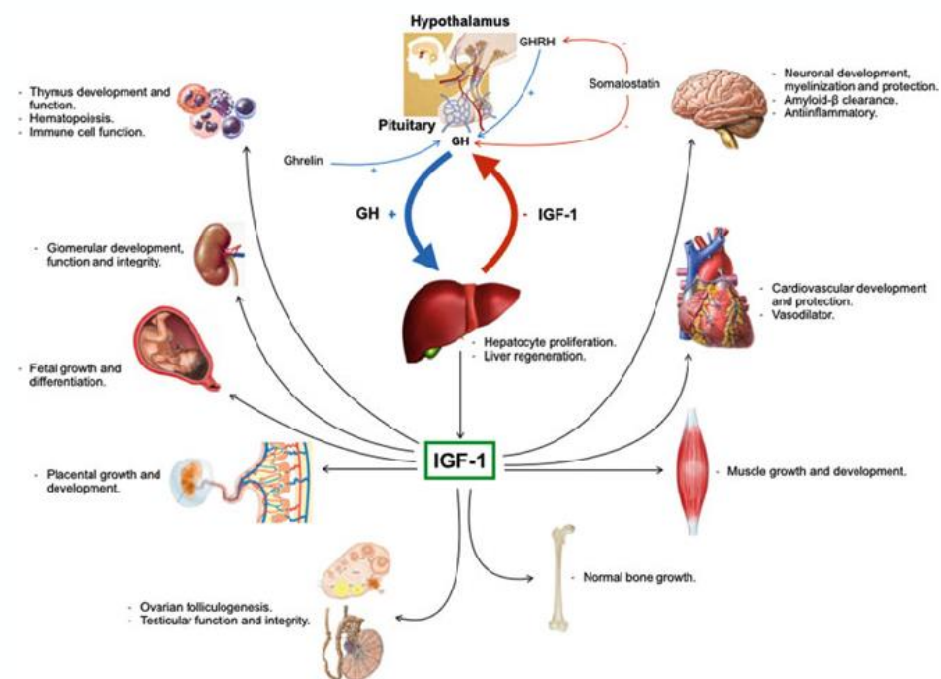
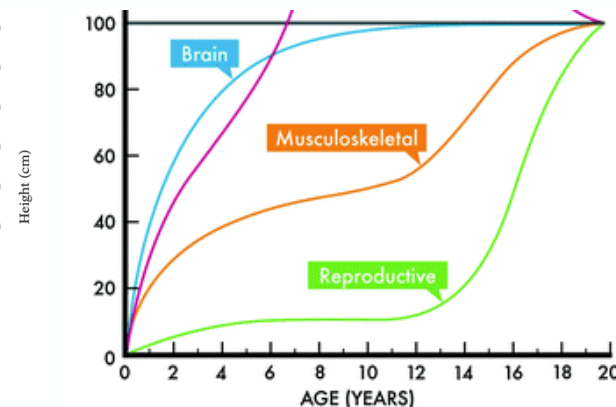
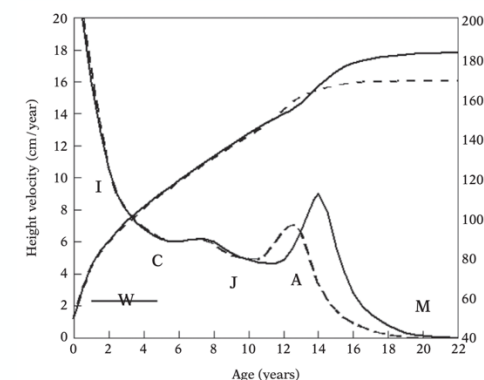
Childhood & juv.

Adol.

Childhood and juvenile growth

Uterine

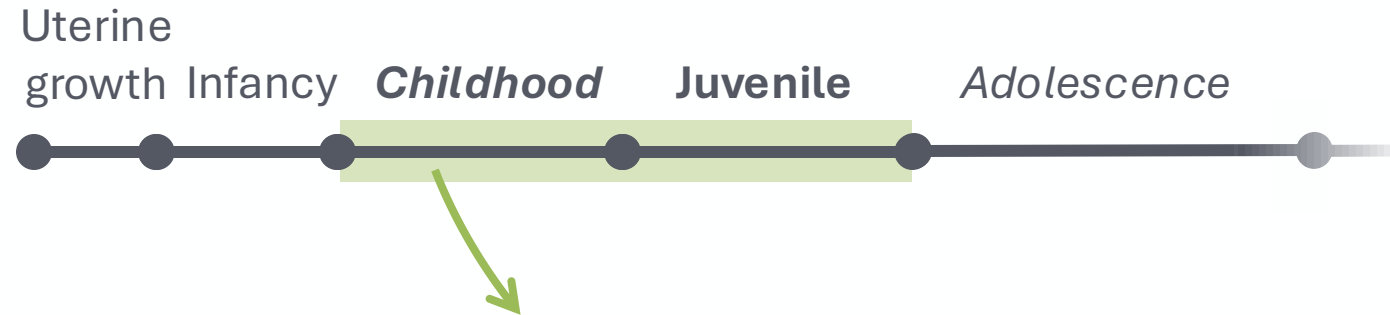
growth Infancy Childhood Juvenile Adolescence



- Linear growth during childhood & juvenile periods: primarily driven by **insulin-like growth factors (IGFs)** and **insulin + growth hormone-IGF1 endocrine axis**

Less flexibility
Due to longer
cascade of hormones
→ more canalized

Childhood & juvenile growth

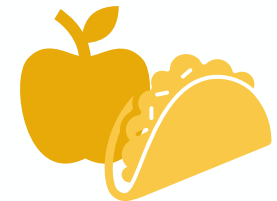


Body growth rates responsive to the environment, but overall rate of growth much lower than in preceding stages

How?

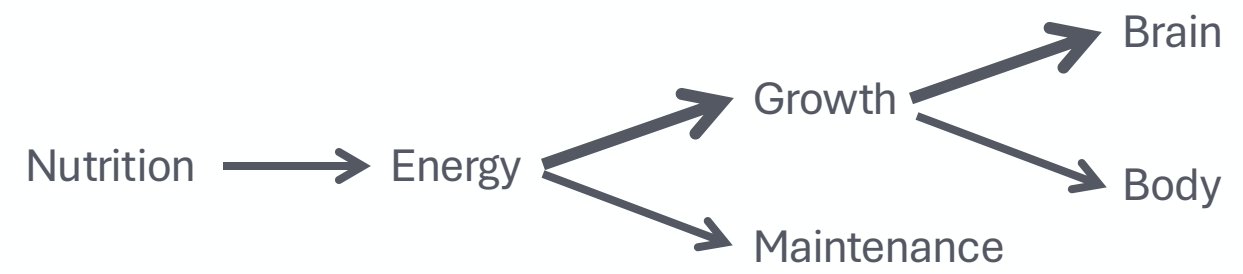
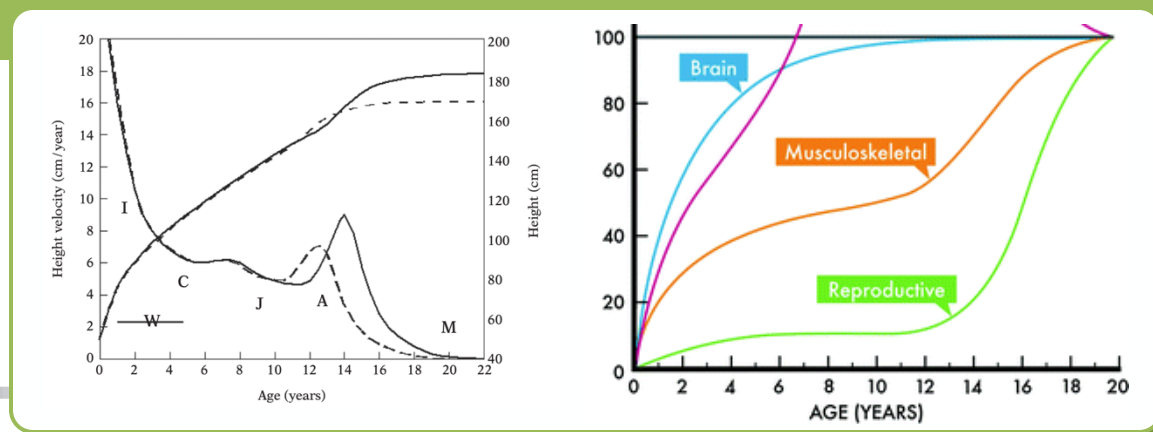
Nutrition remains key: provides energy. Infection & other drivers act by driving energy investment away from growth or affecting energy intake

Linear growth during childhood & juvenile periods: primarily driven by **insulin-like growth factors (IGFs)** and **insulin** + **growth hormone-IGF1 endocrine axis**



Less responsive to quick changes of glucose availability due to longer cascade of hormones

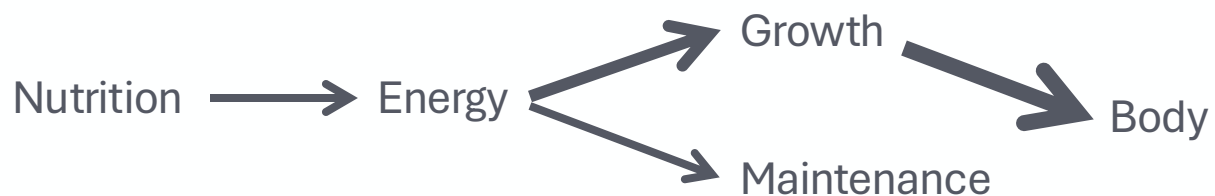
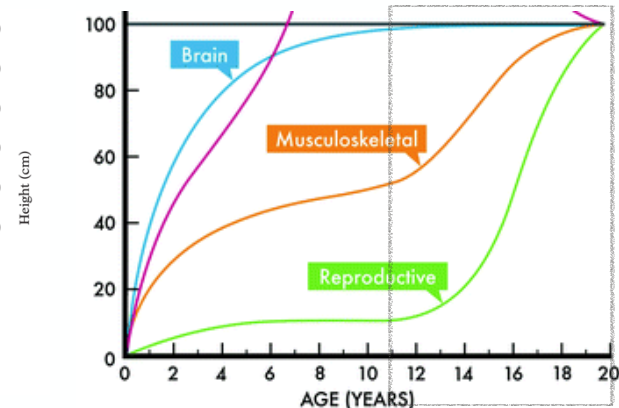
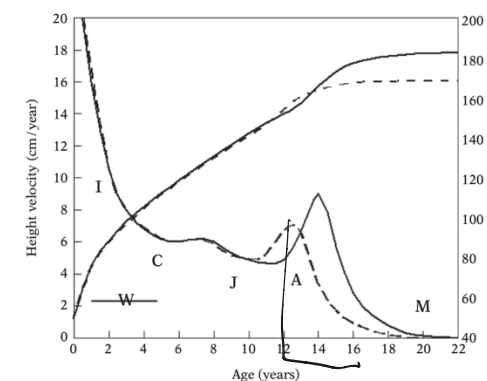
Adolescent growth



Adolescent growth

Uterine

growth Infancy Childhood Juvenile Adolescence



Growth SPURT once
brain g.rowth stops

anemwork

Pre-, perinatal

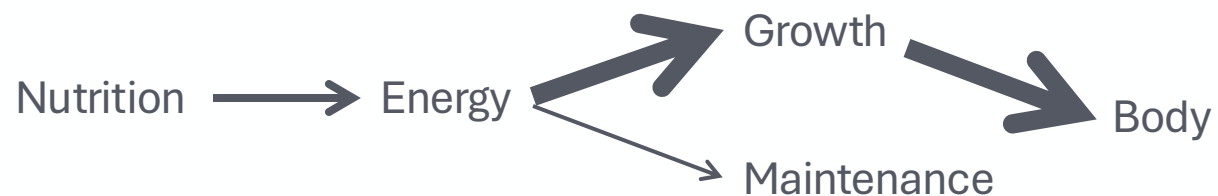
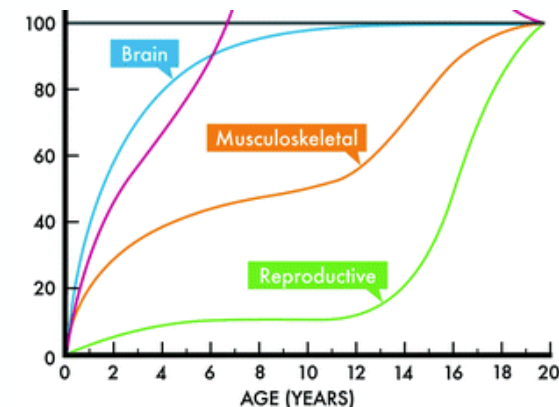
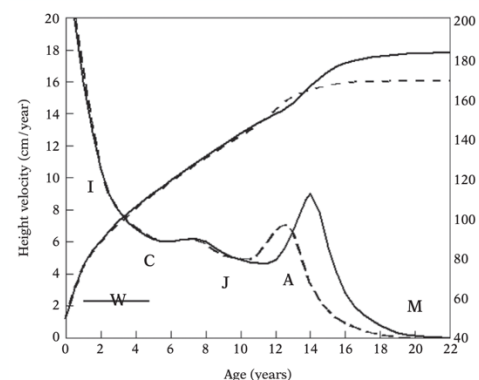
Childhood & juv.

Adol.

Adolescent growth

Uterine

growth Infancy *Childhood* Juvenile *Adolescence*



Mature immune system:
lower disease load

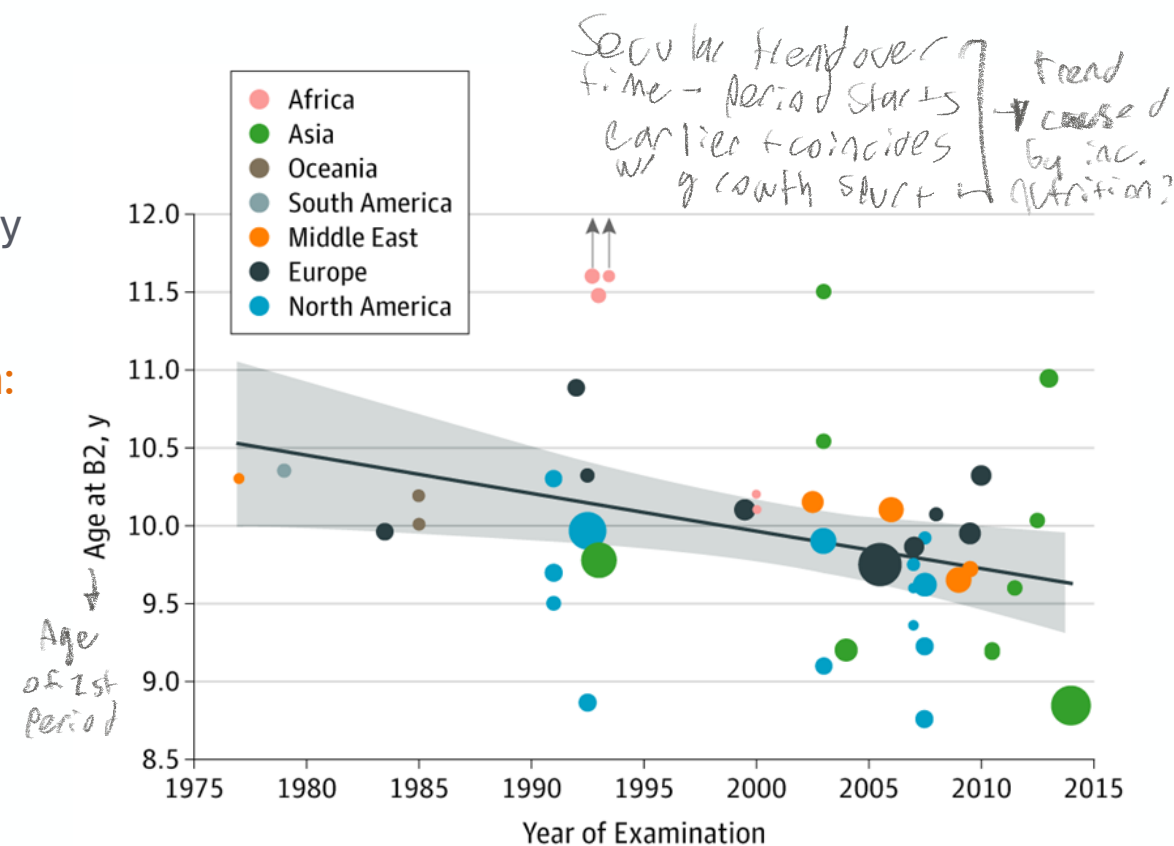
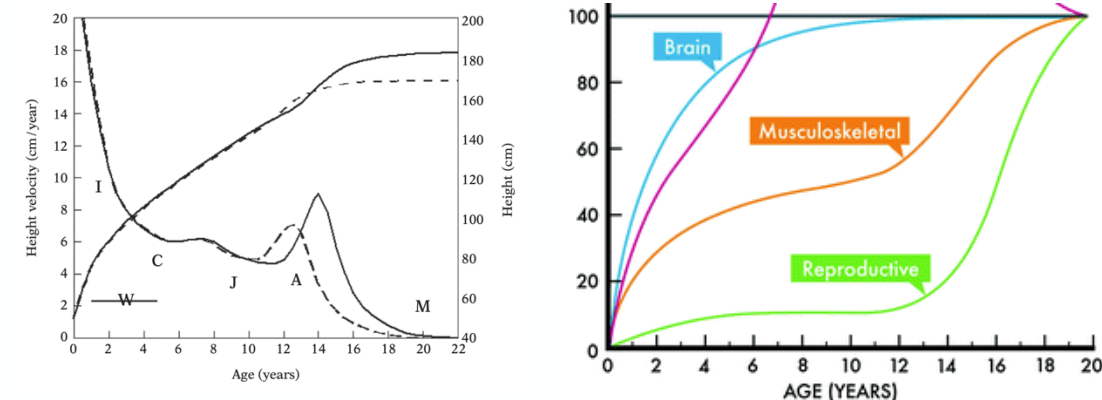
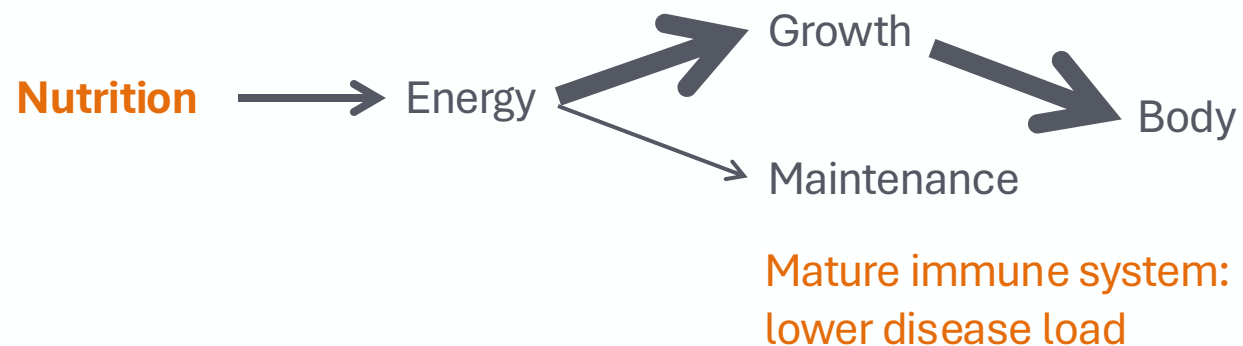
anetwork

Pre-, perinatal

Childhood & juv.

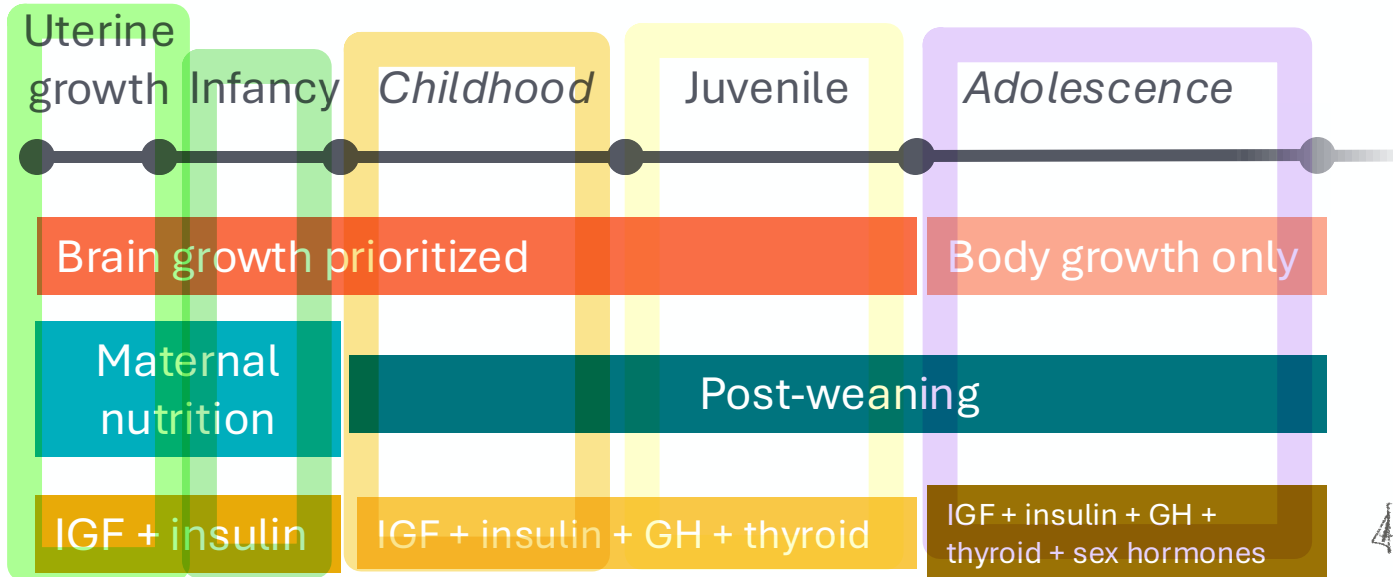
Adol.

Uterine growth Infancy *Childhood* Juvenile *Adolescence*



Summary

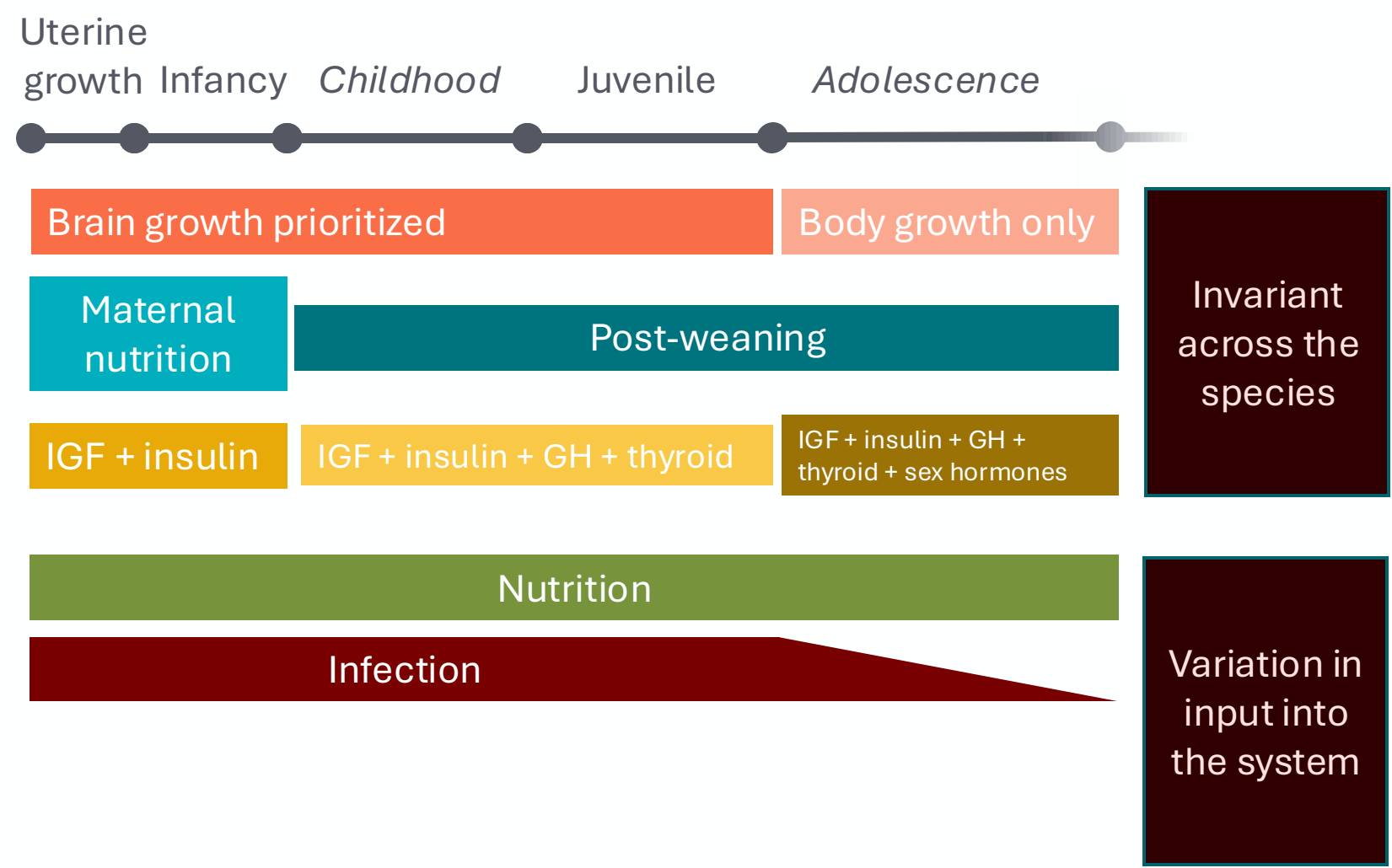
Goal 1: Understand biological regulation of growth in different life history phases



↑ More hormones,
less flexibility
in how size
of growth
responds

↓ Decreasing
flexibility
in response to
nutrition

Summary



Goal 1: Understand biological regulation of growth in different life history phases

Goal 2a: Compare importance of biocultural factors across human development

Goal 2b: ...and between traits

What varies is input into system causing development
 ↳ Nutrition → energy → growth → body
 ↳ brain → maintenance

Summary

Uterine



Brain growth prioritized

Body growth only

Maternal
nutrition

Post-weaning

Invariant
across the
species

IGF + insulin

IGF + insulin + GH + thyroid

IGF + insulin + GH +
thyroid + sex hormones

Nutrition

Infection

Variation in
input into
the system

Poverty, sanitation, housing,
psychosocial stress...

Underlies access to nutrition + susceptibility to disease

**Goal 1: Understand
biological regulation
of growth in different
life history phases**

**Goal 2a: Compare
importance of
biocultural factors
across human
development**

**Goal 2b: ...and
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Summary

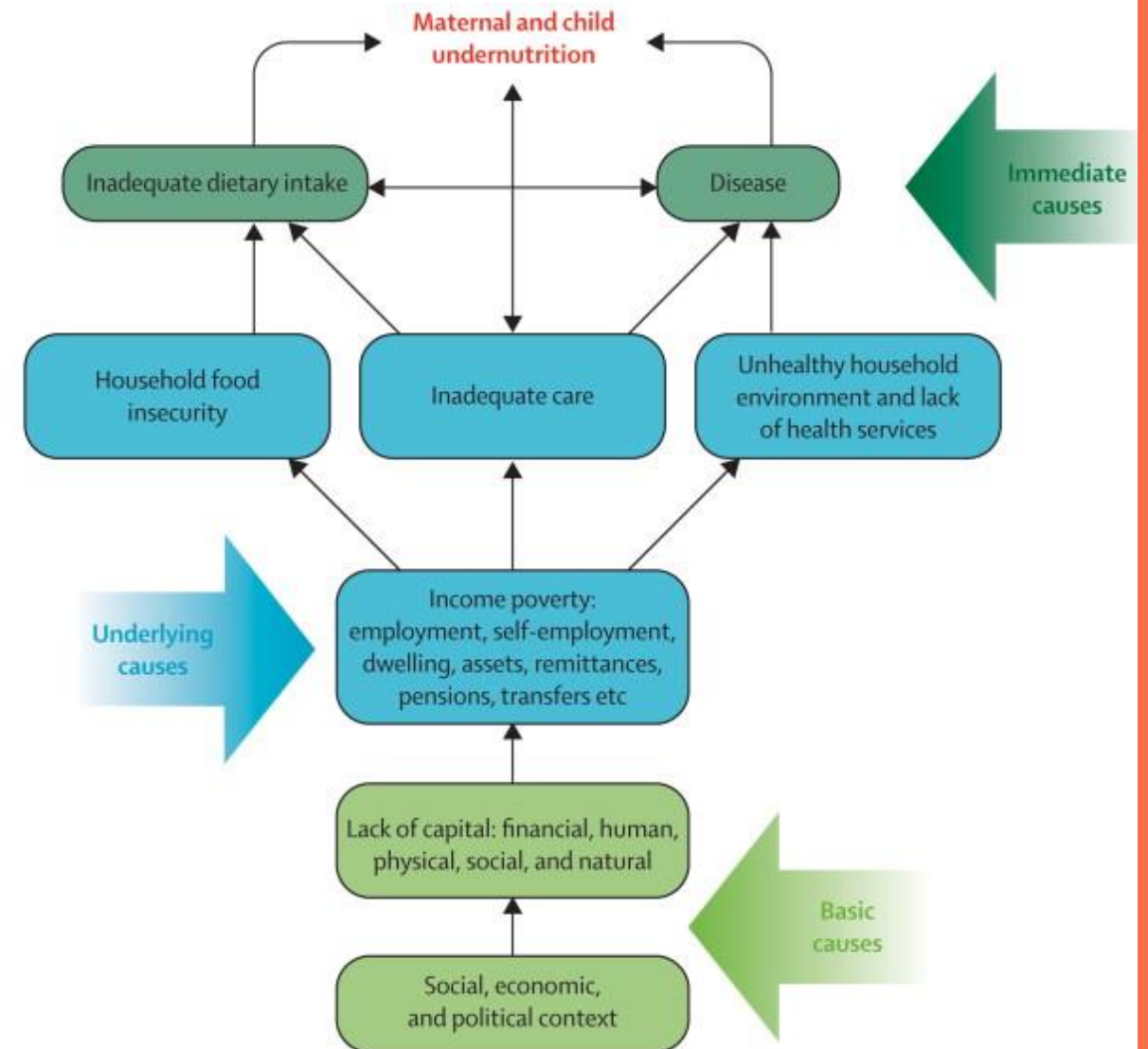
↓ Core pipeline



Poverty, sanitation, housing, psychosocial stress...

A more nuanced answer to 'what is important?':

- Nutrition: simplest pathway to growth
- Disease:
 - immune response competes with growth for energy allocation
 - can affect nutrition (reduced nutrient uptake)
- Cultural context affects nutrition & disease risk
 - more complex relationship with growth, but crucial



Key

Evolution

Framework

Pre-, perinatal

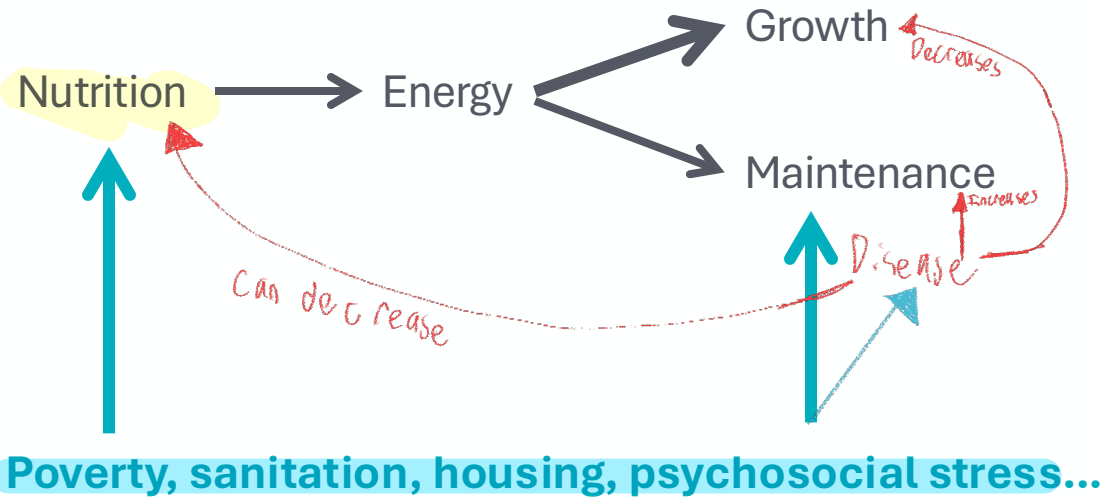
Childhood & juv.

Adol.

Summary

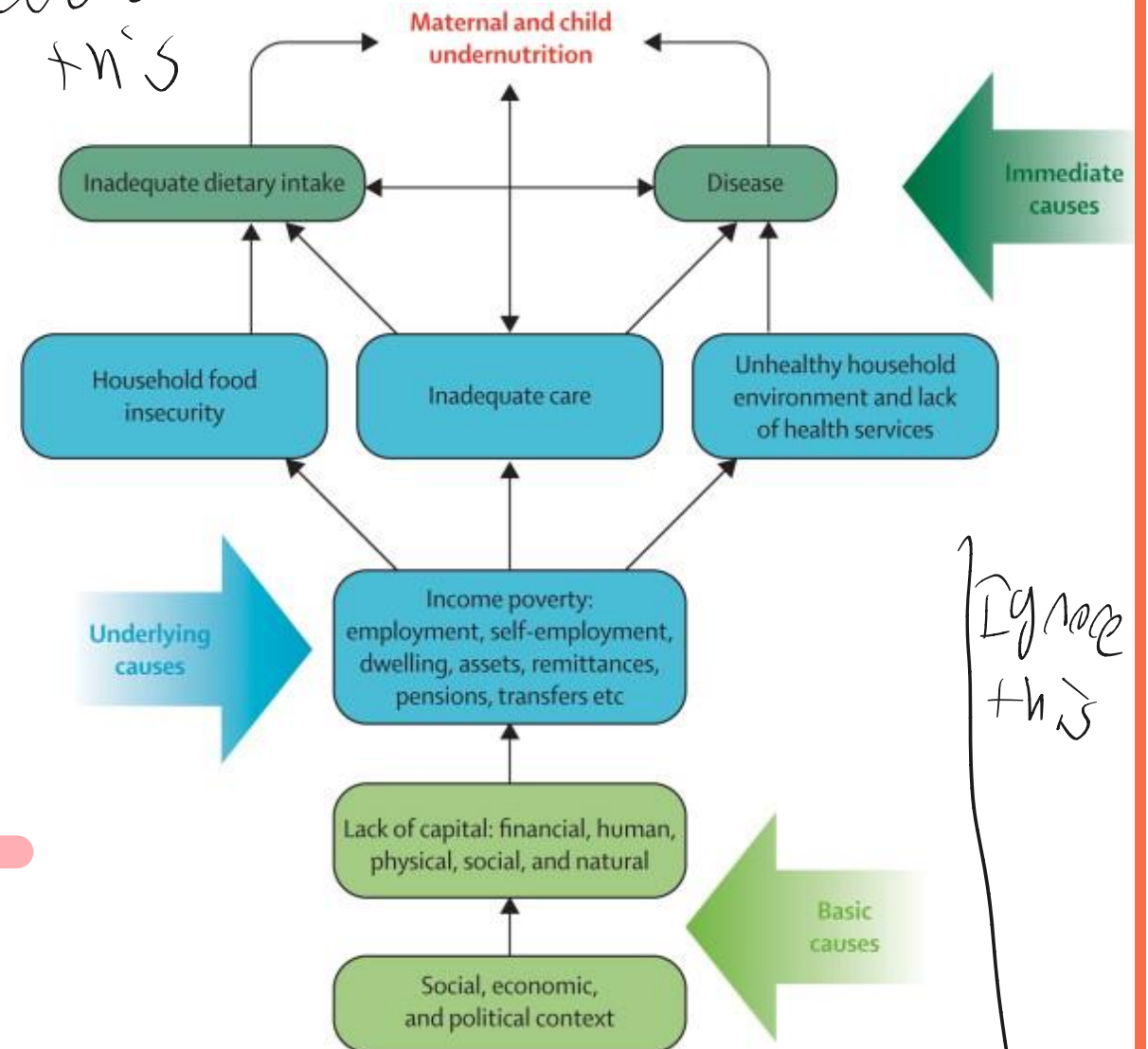
Key

↓ core pipeline

focus on
+n's

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Evolution

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